
DTU

Mathematical Modelling 02526

LDA and Scientific Writing

Welcome to 02526 – Mathematical Modelling

- Course responsibles:
  - ~~Siavash Bigdeli~~
  - ~~Anders Nymark Christensen~~
  - Dimitrios Papadopoulos (Course Responsible)
  - Mirza Karamehmedovic (Course co-responsible)

- The course is 13 weeks - each Tuesday from 08.00 – 12.00
- Three exercise projects with hand in of reports plus peergrading
- One exam project and report
- Help during exercises projects each Tuesday

Last week

- Learning objectives
- Practicalities: Course Plan, Evaluation and exams
- Introduction to mathematical modeling
- Exercise 1: Salami Quality

Course Plan

- The course is fully physical
- 13 weeks, Tuesday from 08.00 – 12.00
- 5 ECTS work load is ~9 hours per week (including the 4 hours each Tuesday)
- 3 Exercises projects run over 3 weeks
- Hand in of reports plus peergrading
- 1 Exam project an report over 4 weeks
- The work is done in groups of 3-4 people. Pairs can be acceptable, but solo is not recommended
- Report hand-in on Sunday before the next exercise
- **Introduction and follow-up lecture for each project**
- **You will read each other's report(s) and comment on them through Peergrade**
  - **Peergrading is mandatory, and a hand in will count for nothing if not done**

Date	What	Who
Tuesday 3/2	Welcome + Salami introduction	Dimitrios, Mirza
Tuesday 10/2	LDA + Scientific writing	Dimitrios
Tuesday 17/2	Peergrade and feedback	Dimitrios
Sunday 22/2	HAND IN - Salami	
Tuesday 24/2	Optical flow intro	Mirza
Sunday 1/3	Deadline - FeedbackFruit	
Tuesday 3/3	Optical flow	Mirza
Tuesday 10/3	Optical flow + Salami feedback	Mirza, Dimitrios
Sunday 15/3	HAND IN - Optical flow	
Tuesday 17/3	Integer programming	Stefan, Dimitrios
Sunday 22/3	Deadline - FeedbackFruit	
Tuesday 24/3	Exercise followup	Mirza
Tuesday 31/3	Easter break	
Sunday 5/4	HAND IN - Integer programming	
Tuesday 7/4	Optical flow feedback	Mirza
Tuesday 14/4	Exam project Q&A, Integer Programming feedback	All
Sunday 19/4	Deadline - FeedbackFruit	
Tuesday 21/4	Exam project	
Tuesday 28/4	Exam project	
Tuesday 5/5	Exam project	
Sunday 10/5	HAND IN - Exam project	
Thursday 21/5	Oral Exams	

Exam

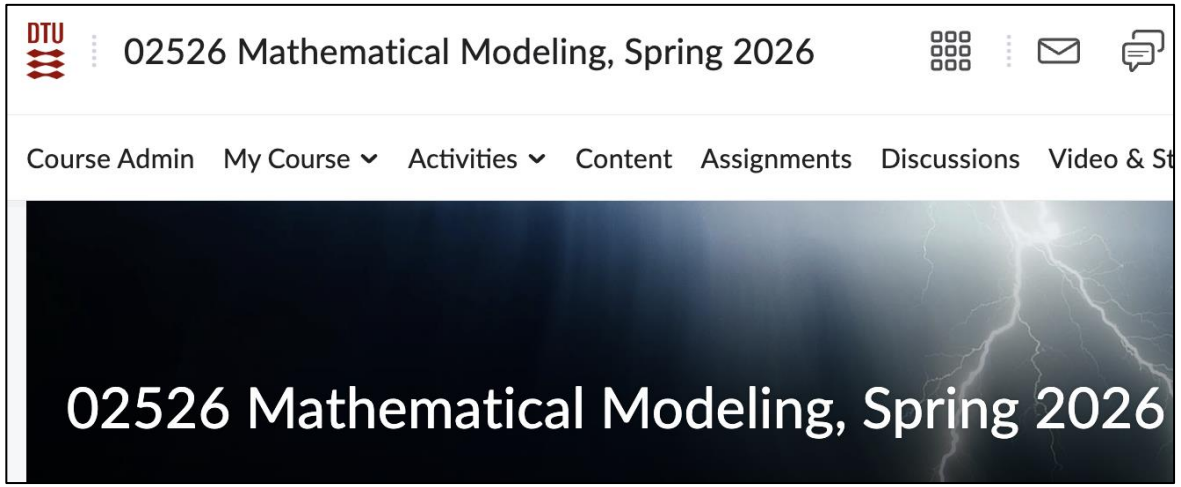
- The exercise reports (and peergrading) are pass/no pass.
  - Formative feedback given
  - *Must pass all exercise projects to pass the course*

- Exam project run over 4 weeks
  - Largely unassisted

- Short oral defense with the basis in the project report
  - Summative feedback
  - External examiner
  - Exam dates are yet to be settled

Useful Information, platforms and communication

DTU Learn



https://learn.inside.dtu.dk/d2l/home/296292

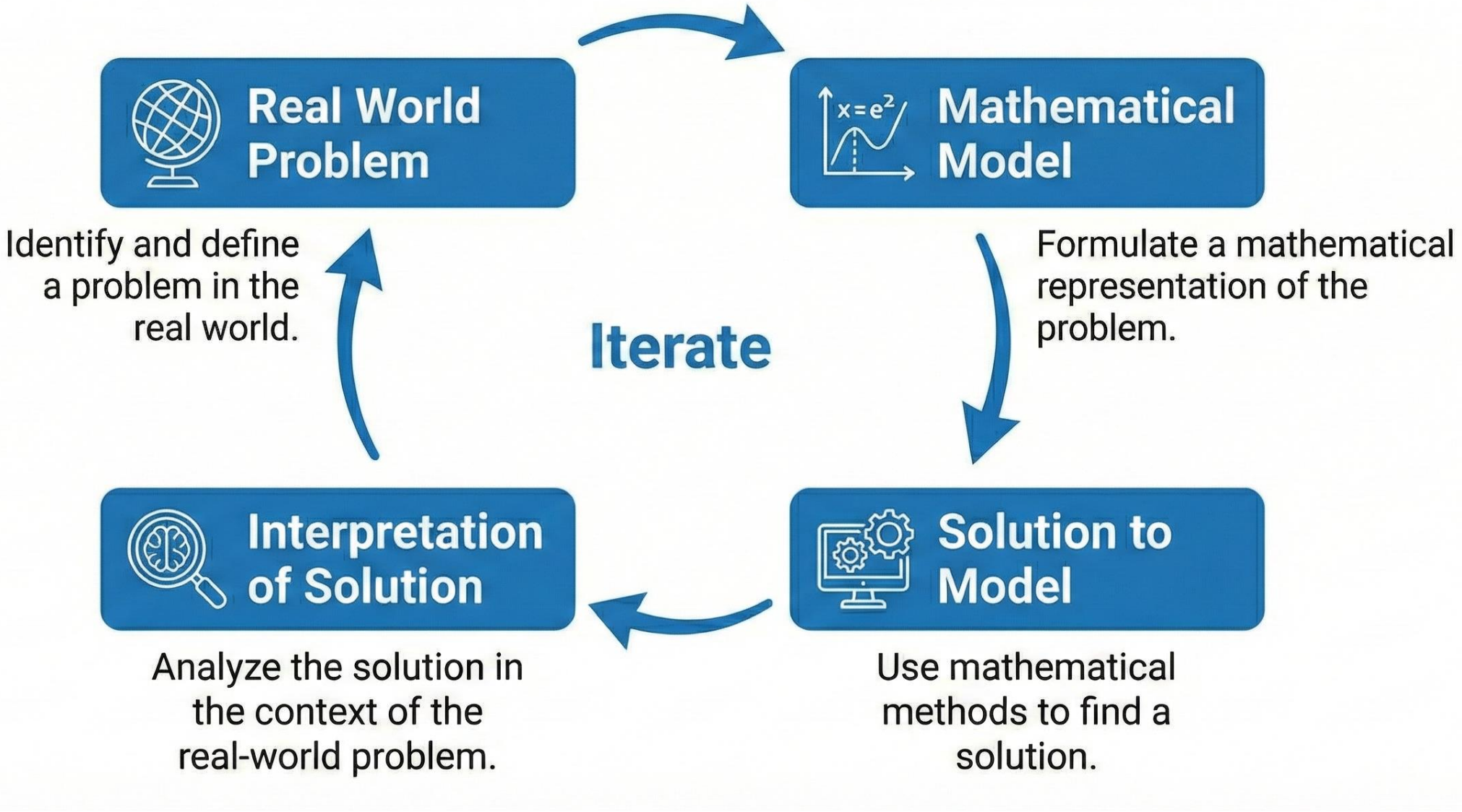
- All material (or links)
- Curriculum
- Lectures
- Exercises
- Handins/ Peergrade

Slack workspace

https://join.slack.com/t/02526mathemat-0wh5248/shared_invite/zt-3pferlik0-uiJ0PE37ds1sUwZv2bstqQ

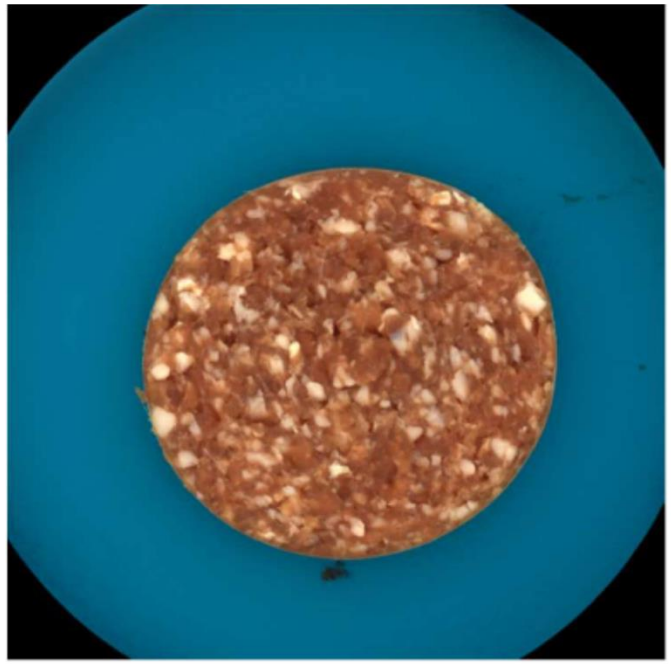
- Group members
- Questions for exercises
- General questions
- Announcements

The modeling cycle

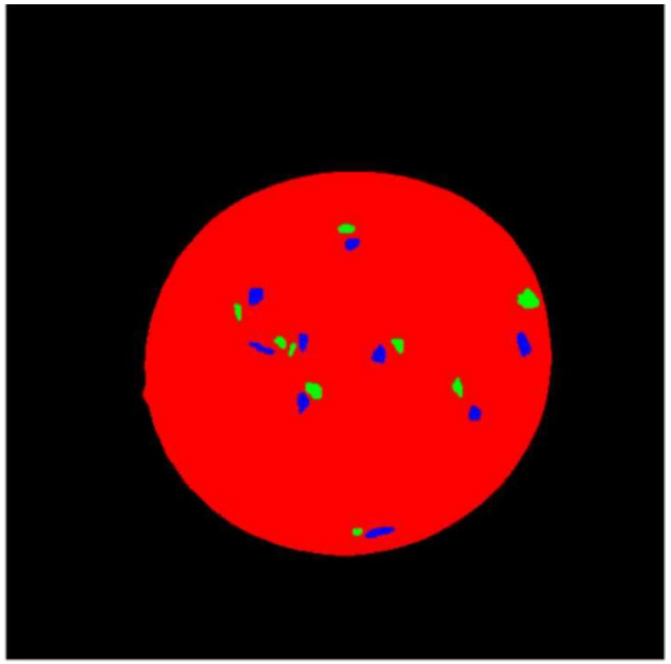


Mathematical model

- **Problem**: Classify pixels to meat/fat
- **Training data**: known labels for pixels
- **Classifcation**: Supervised machine learning algorithm



Day 1

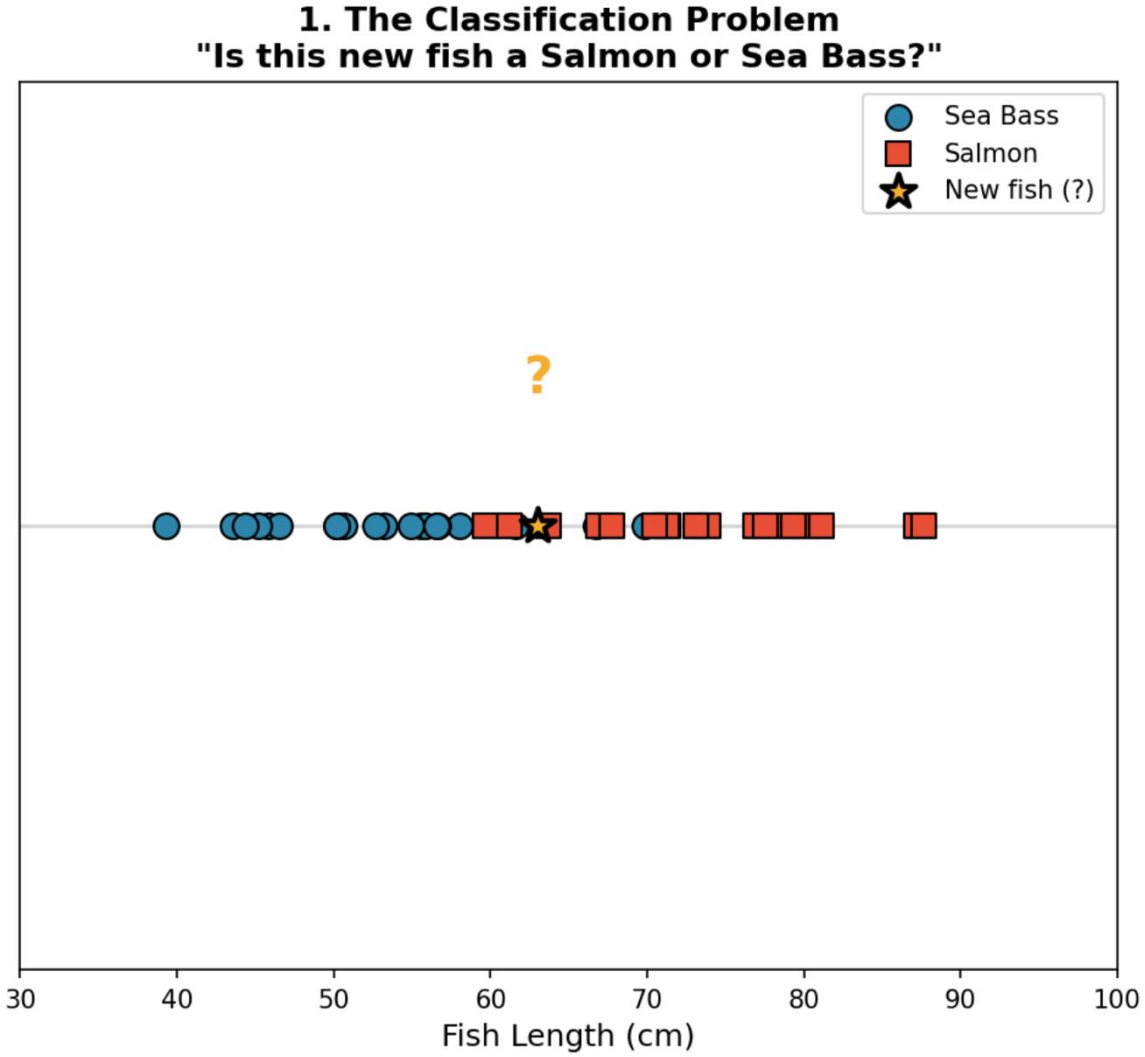


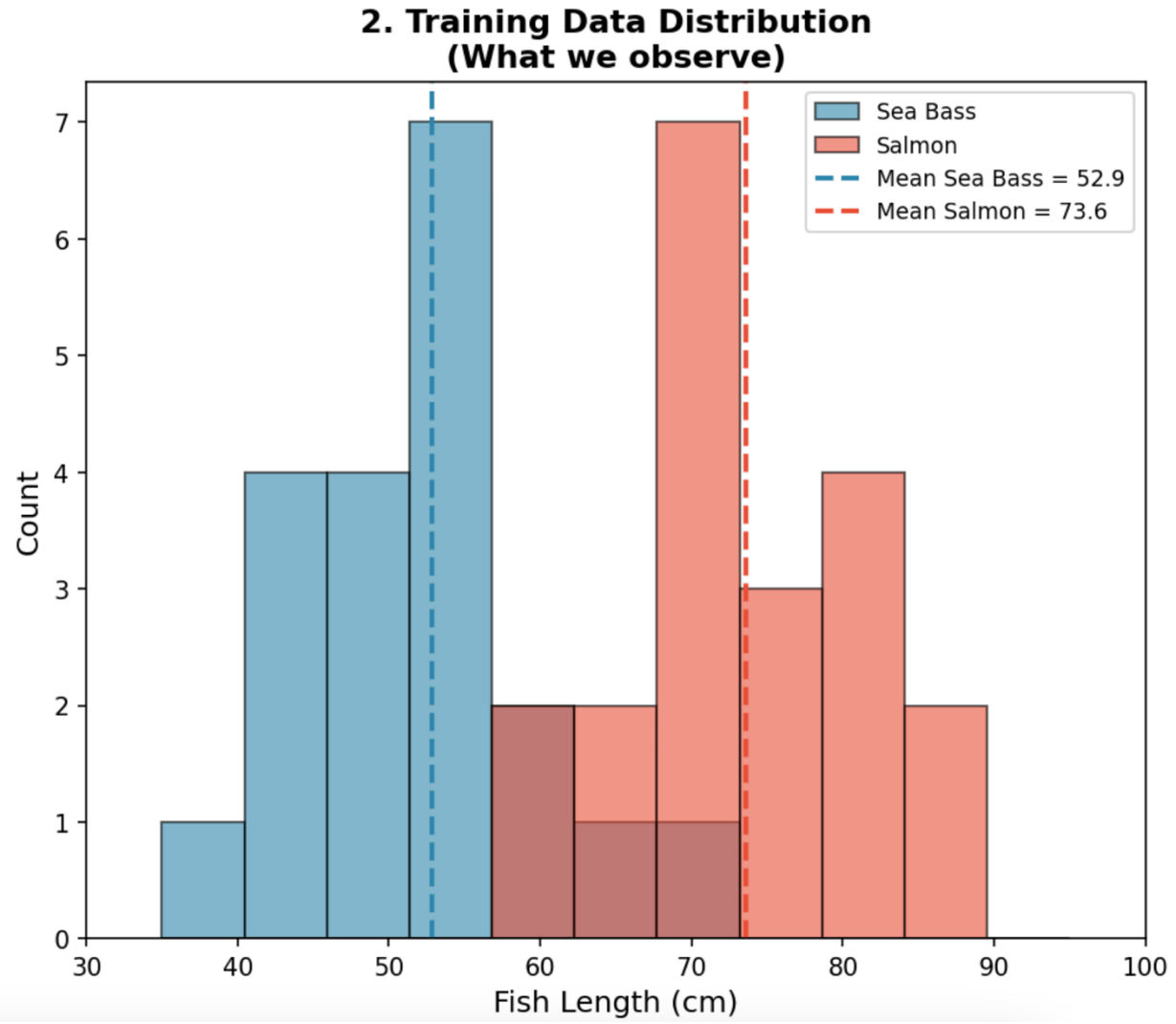
Annotation

Today's Lecture

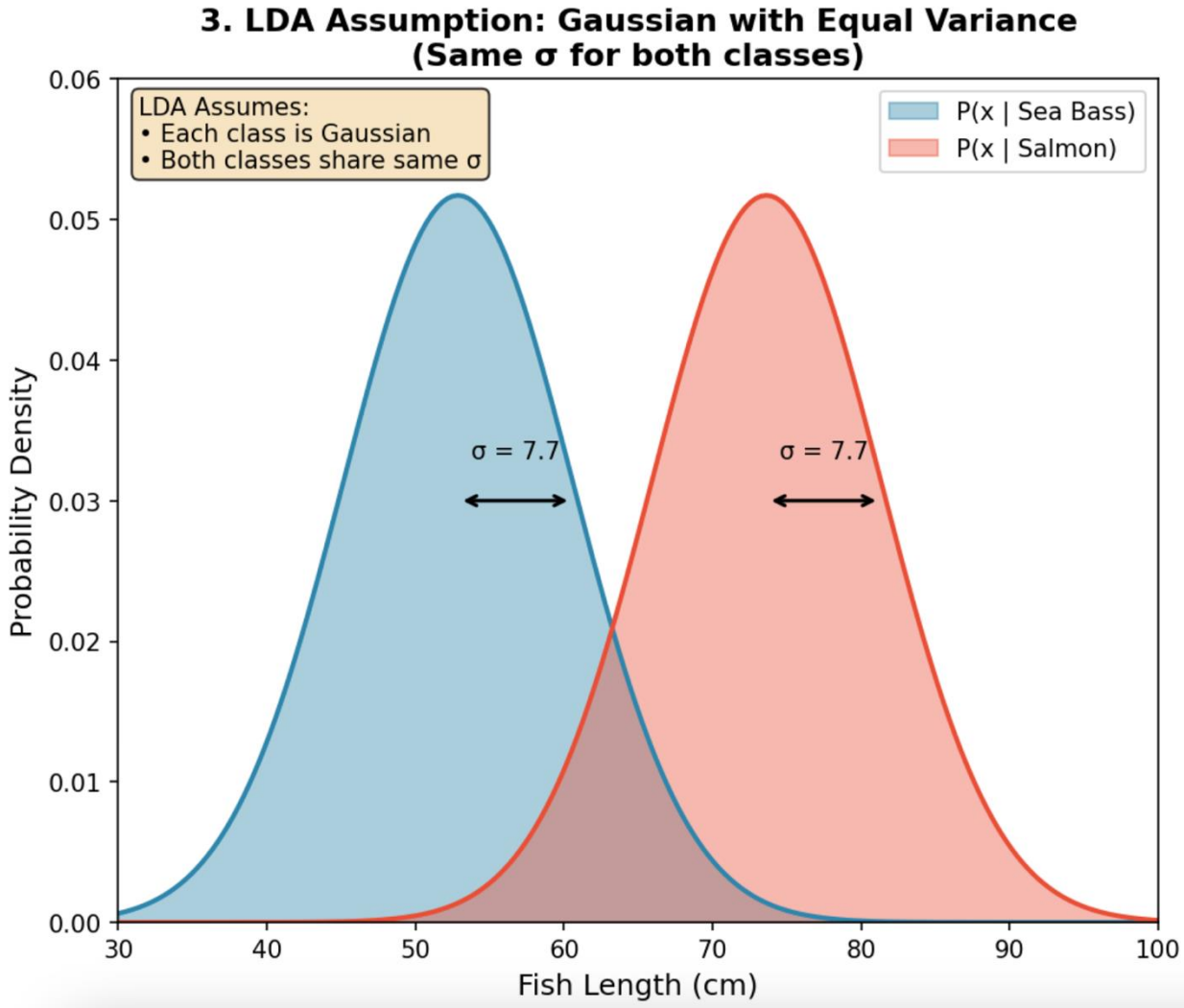
- LDA and Exercise 1
- Scientific Writing
- Use of AI tools

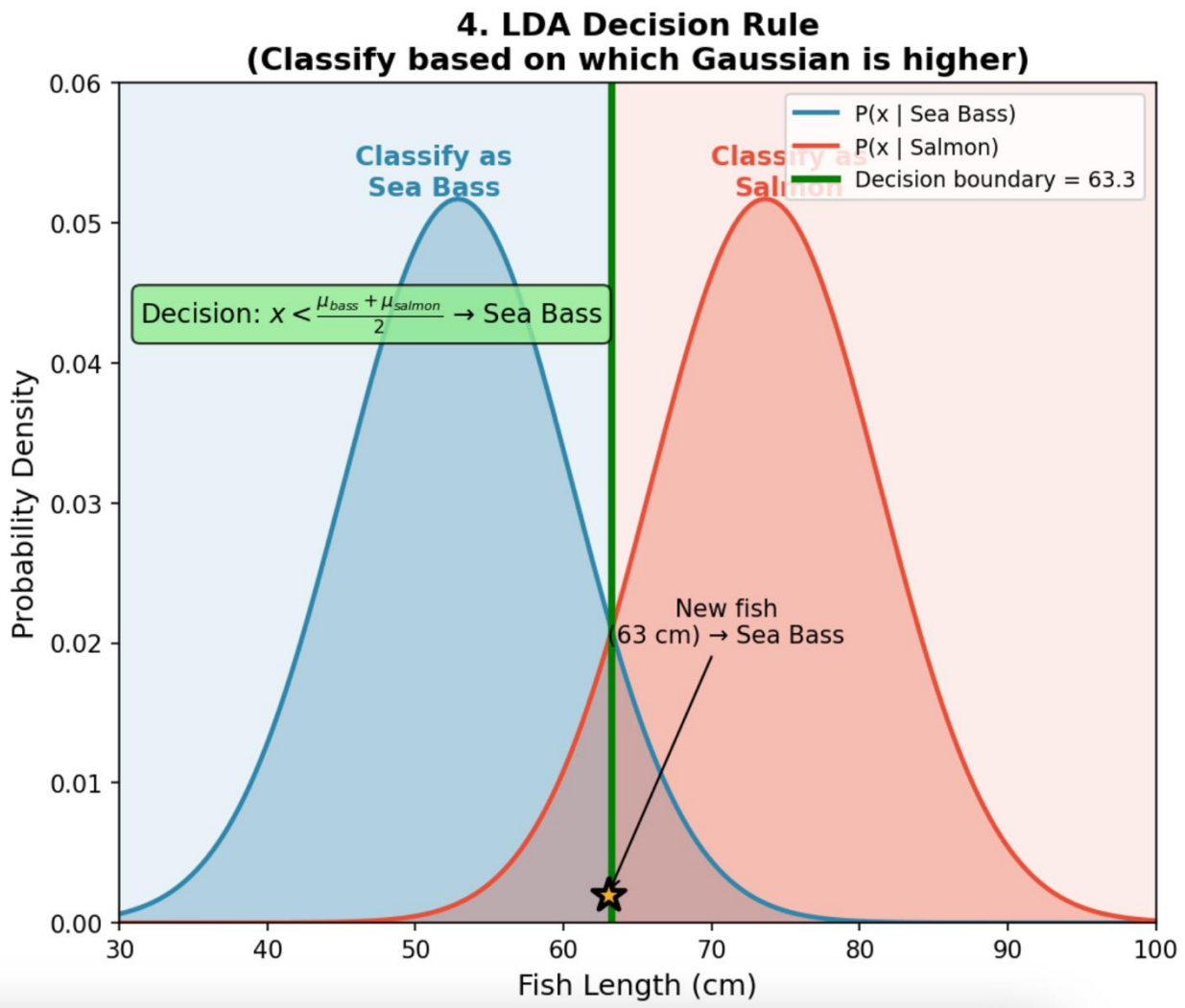
Linear Discriminant Analysis





Linear Discriminant Analysis





Linear Discriminant Analysis

Problem and Assumptions

Sea Bass: $\mathcal{D}_1 = \{x_1^{(1)}, x_2^{(1)}, \ldots, x_{n_1}^{(1)}\}$ where $n_1 = 20$

Salmon: $\mathcal{D}_2 = \{x_1^{(2)}, x_2^{(2)}, \ldots, x_{n_2}^{(2)}\}$ where $n_2 = 20$

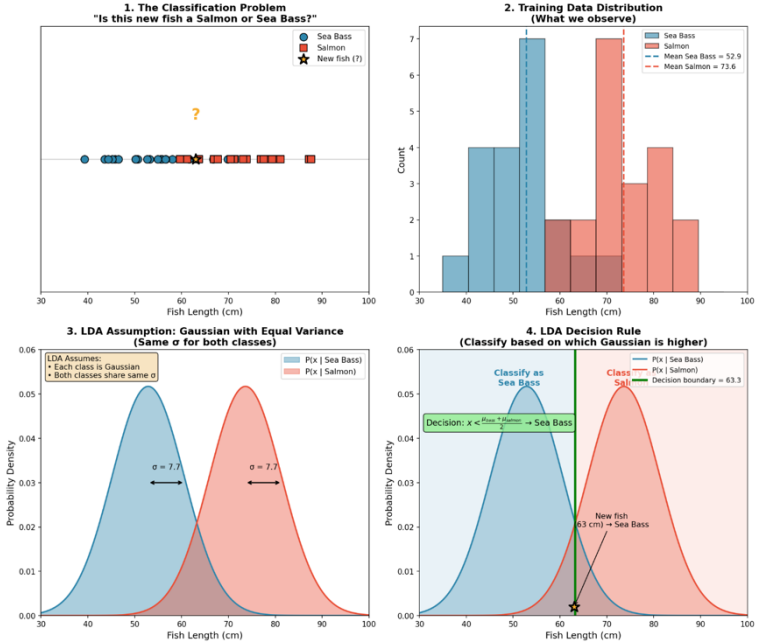
$$P(x \mid \text{Sea Bass}) = \mathcal{N}(x \mid \mu_1, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu_1)^2}{2\sigma^2}\right)$$

$$P(x \mid \text{Salmon}) = \mathcal{N}(x \mid \mu_2, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu_2)^2}{2\sigma^2}\right)$$

find $P(\text{class} \mid x^*)$

Key assumption: Both classes share the same σ^2 (homoscedasticity)

Prior probabilities: $\pi_1 = P(\text{Sea Bass})$ and $\pi_2 = P(\text{Salmon})$, with $\pi_1 + \pi_2 = 1$



Linear Discriminant Analysis

Problem and Assumptions

Sea Bass: $\mathcal{D}_1 = \{x_1^{(1)}, x_2^{(1)}, \ldots, x_{n_1}^{(1)}\}$ where $n_1 = 20$

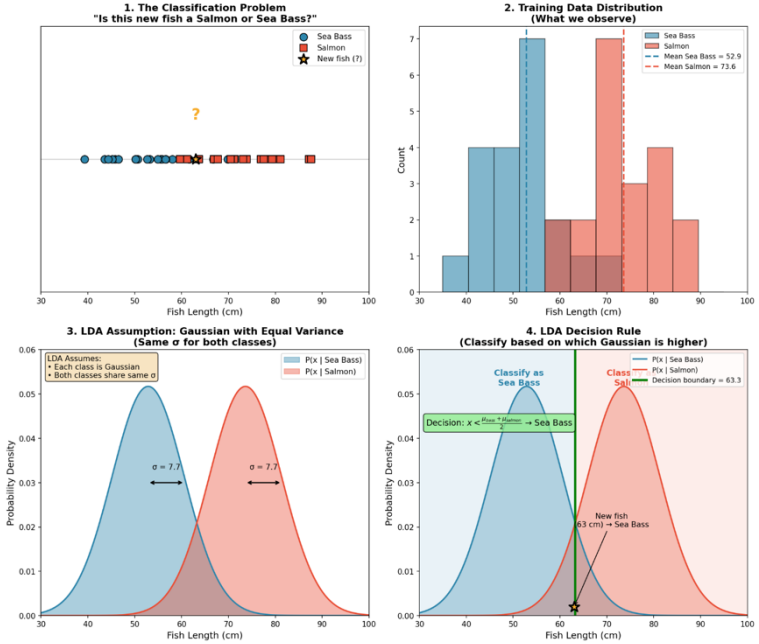
Salmon: $\mathcal{D}_2 = \{x_1^{(2)}, x_2^{(2)}, \ldots, x_{n_2}^{(2)}\}$ where $n_2 = 20$

$$find\ P(\text{class} \mid x^*)$$

$$P(x \mid \text{Sea Bass}) = \mathcal{N}(x \mid \mu_1, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu_1)^2}{2\sigma^2}\right)$$

$$P(x \mid \text{Salmon}) = \mathcal{N}(x \mid \mu_2, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu_2)^2}{2\sigma^2}\right)$$

Key assumption: Both classes share the same σ^2 (homoscedasticity)



Estimate Parameters from training data

Prior probabilities: $\pi_1 = P(\text{Sea Bass})$ and $\pi_2 = P(\text{Salmon})$, with $\pi_1 + \pi_2 = 1$

Class means (Maximum Likelihood Estimates):

$$\hat{\mu}_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} x_i^{(1)} = 52.87 \text{ cm} \quad \text{(Sea Bass)}$$

$$\hat{\mu}_2 = \frac{1}{n_2} \sum_{i=1}^{n_2} x_i^{(2)} = 73.63 \text{ cm} \quad \text{(Salmon)}$$

Pooled variance estimate (using both classes):

$$\hat{\sigma}^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1 + n_2 - 2} = \frac{19 \cdot s_1^2 + 19 \cdot s_2^2}{38}$$

$$\hat{\sigma} = 7.71 \text{ cm}$$

Linear Discriminant Analysis

Apply Bayes' Rule

Posterior probability using Bayes' theorem:

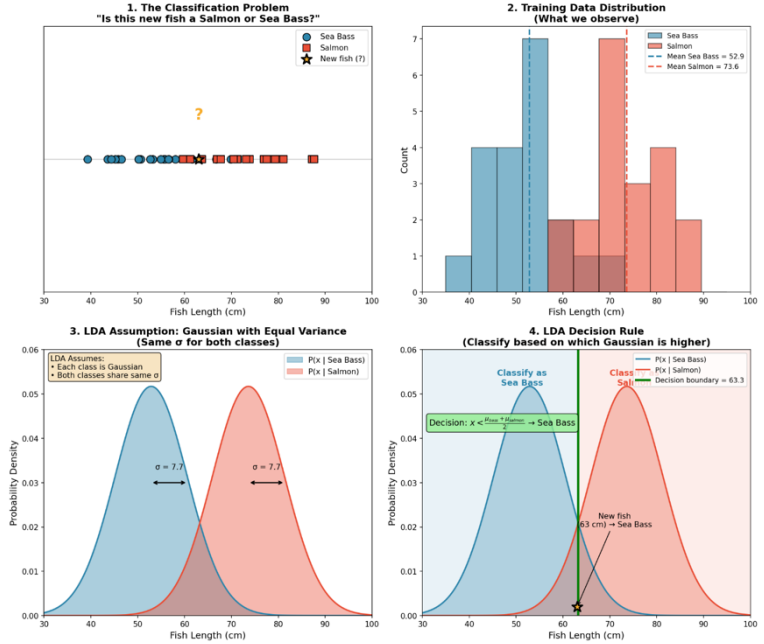
$$P(\text{class}_k \mid x) = \frac{P(x \mid \text{class}_k) \cdot \pi_k}{P(x)} = \frac{P(x \mid \text{class}_k) \cdot \pi_k}{\sum_j P(x \mid \text{class}_j) \cdot \pi_j}$$

Decision rule: Classify to the class with higher posterior. Equivalently, compare:

$$P(x \mid \text{Sea Bass}) \cdot \pi_1 \quad \text{vs} \quad P(x \mid \text{Salmon}) \cdot \pi_2$$

Taking the log ratio (log-odds) and simplifying:

$$\log \frac{P(\text{Sea Bass} \mid x)}{P(\text{Salmon} \mid x)} = \log \frac{\pi_1}{\pi_2} + \frac{\mu_1 - \mu_2}{\sigma^2} x + \frac{\mu_2^2 - \mu_1^2}{2\sigma^2}$$



Linear Discriminant Analysis

Find Decision Boundary Apply Bayes' Rule

Posterior probability using Bayes' theorem:

$$P(\text{class}_k \mid x) = \frac{P(x \mid \text{class}_k) \cdot \pi_k}{P(x)} = \frac{P(x \mid \text{class}_k) \cdot \pi_k}{\sum_j P(x \mid \text{class}_j) \cdot \pi_j}$$

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With equal priors ($\pi_1 = \pi_2 = 0.5$), the decision boundary is where posteriors are equal:

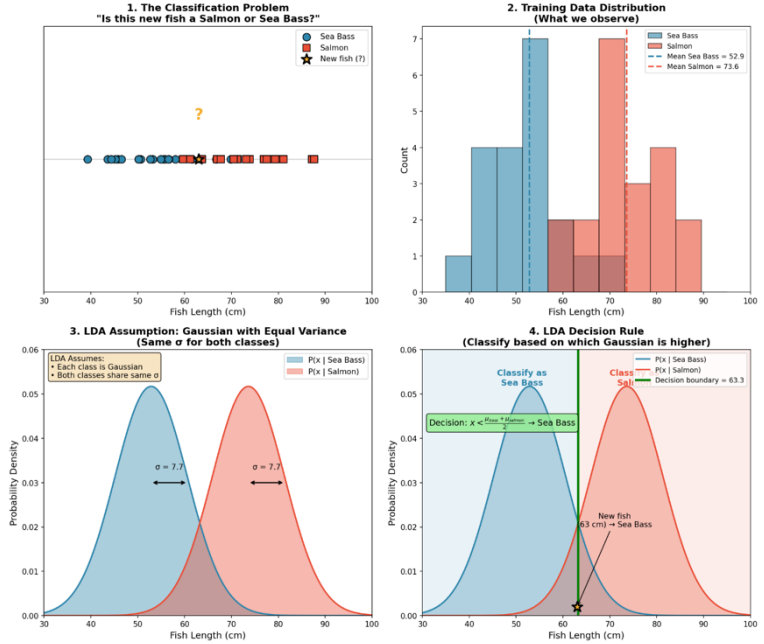
$$P(x \mid \text{Sea Bass}) = P(x \mid \text{Salmon})$$

Solving for x (the Gaussians intersect at):

$$x^* = \frac{\mu_1 + \mu_2}{2} = \frac{52.87 + 73.63}{2} = 63.25 \textbf{ cm}$$

LDA Decision Rule:

If $x < 63.25 \rightarrow$ classify as Sea Bass; If $x \geq 63.25 \rightarrow$ classify as Salmon



Multivariate Linear Discriminant Analysis

$$f_i(\mathbf{x}) = \frac{1}{\sqrt{2\pi}^n \sqrt{\det \hat{\mathbf{\Sigma}}}} \exp\left(-\frac{1}{2}(\mathbf{x} - \hat{\boldsymbol{\mu}}_i)^T \hat{\mathbf{\Sigma}}^{-1}(\mathbf{x} - \hat{\boldsymbol{\mu}}_i)\right)$$

PDF of obsevavation **x**

$$S_i(\mathbf{x}) = \mathbf{x}^T \hat{\mathbf{\Sigma}}^{-1} \hat{\boldsymbol{\mu}}_i - \frac{1}{2} \hat{\boldsymbol{\mu}}_i{}^T \hat{\mathbf{\Sigma}}^{-1} \hat{\boldsymbol{\mu}}_i + \ln p_i$$

Discriminant function S

We classify a pixel **x** to class *i* if $S_i(x)$ is maximized

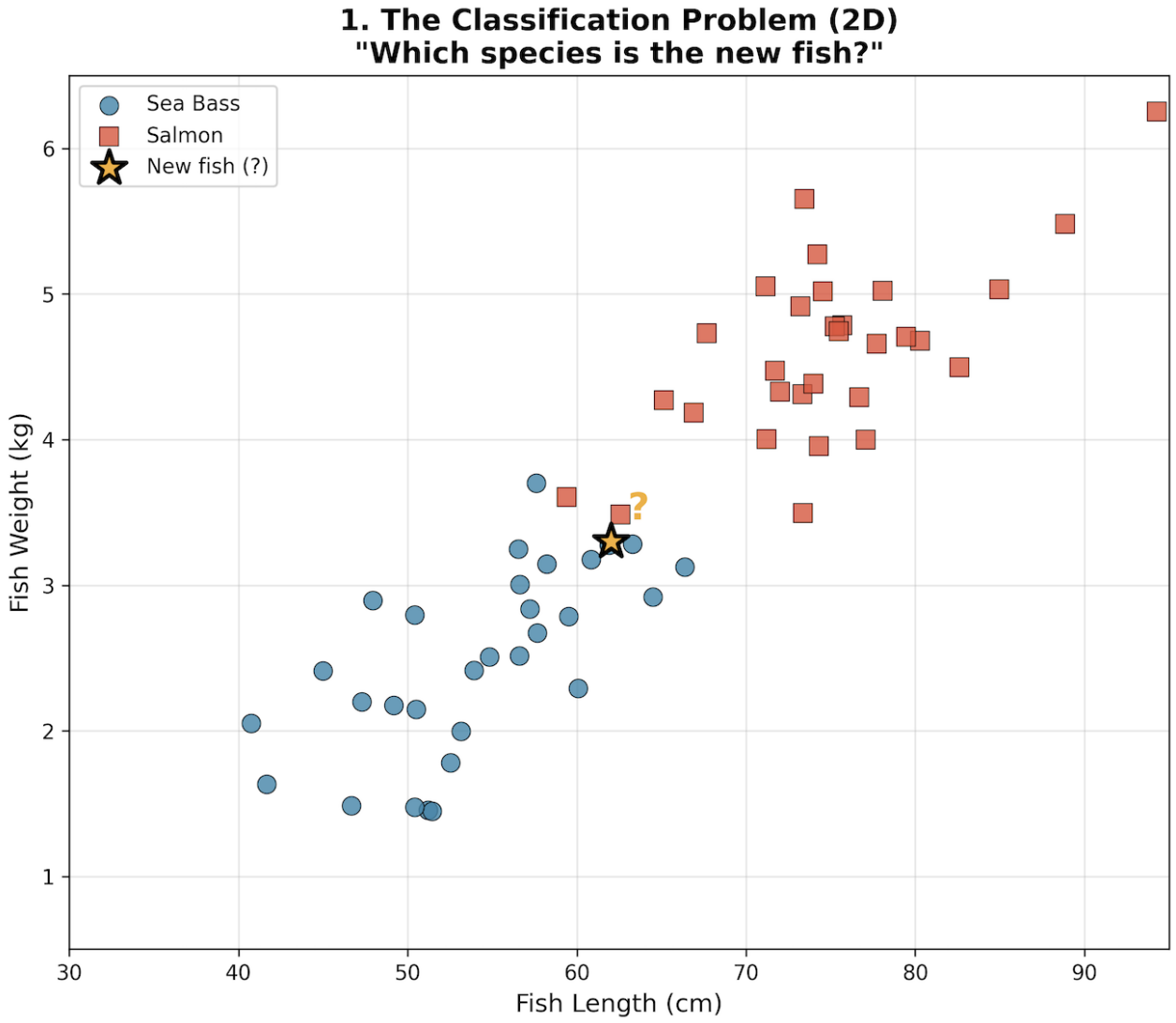
x: The 19-dimensional pixel vector

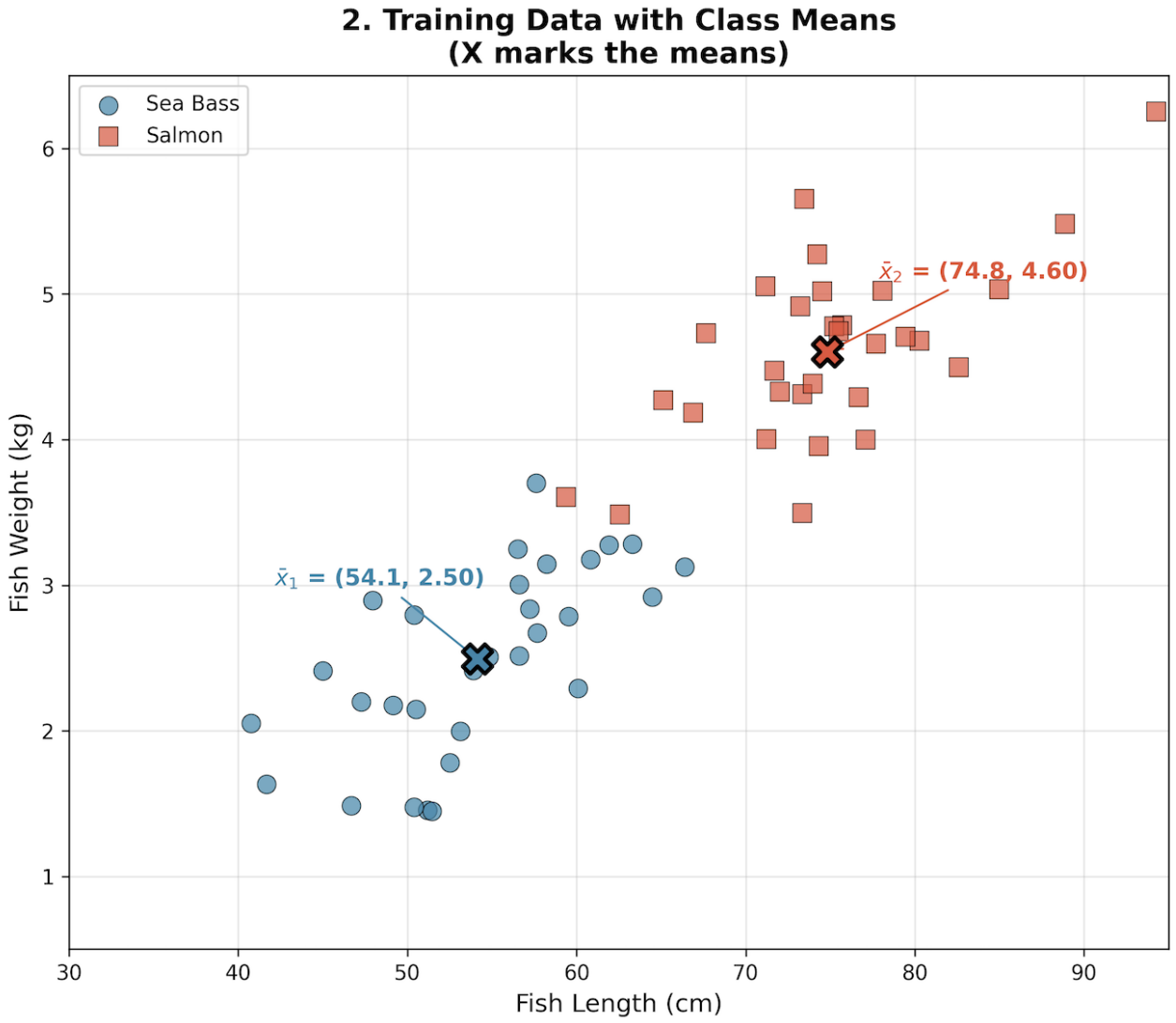
Σ: The pooled Covariance Matrix

μ: The mean spectral vector for class *i*

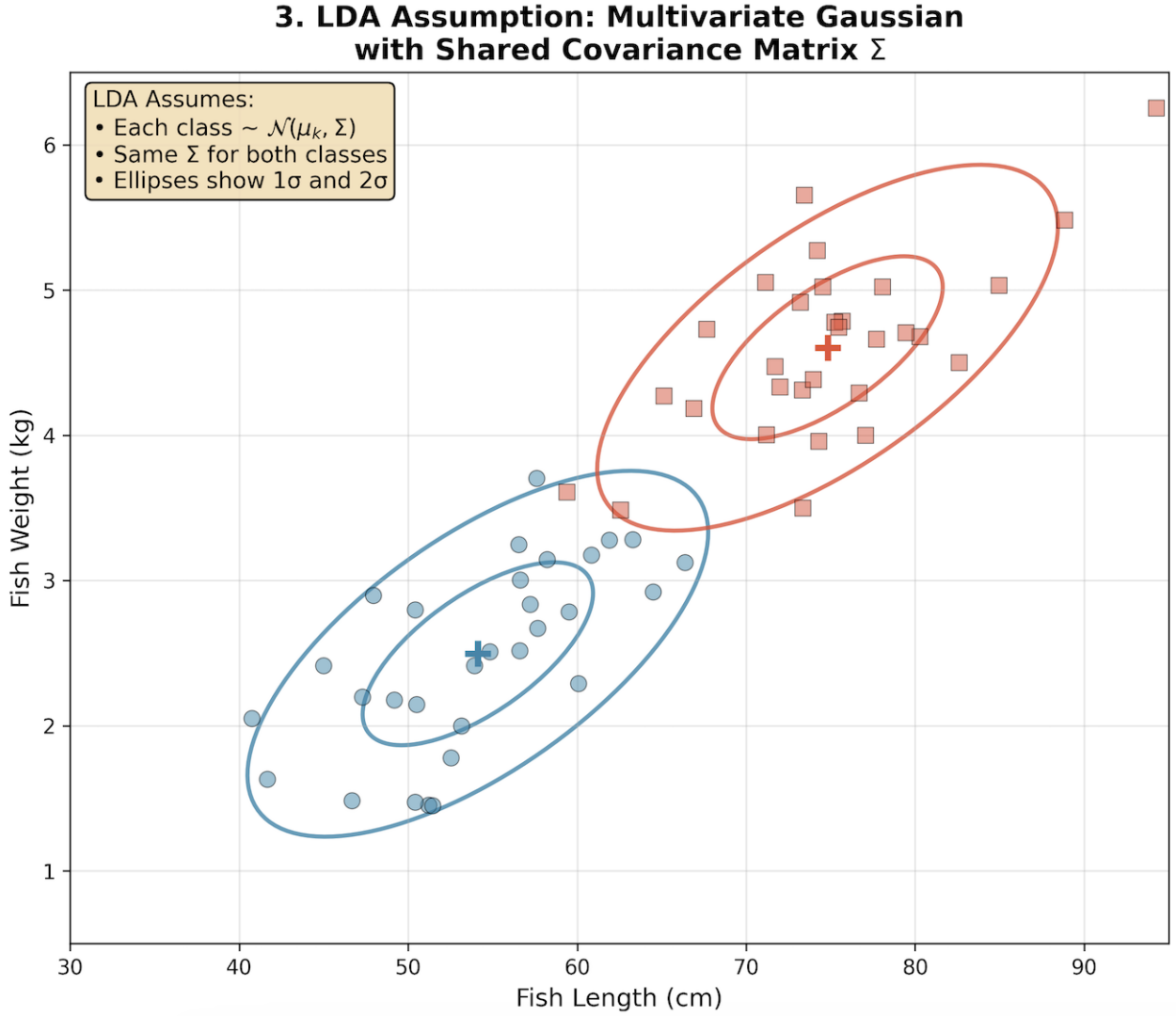
p_i: The prior probability

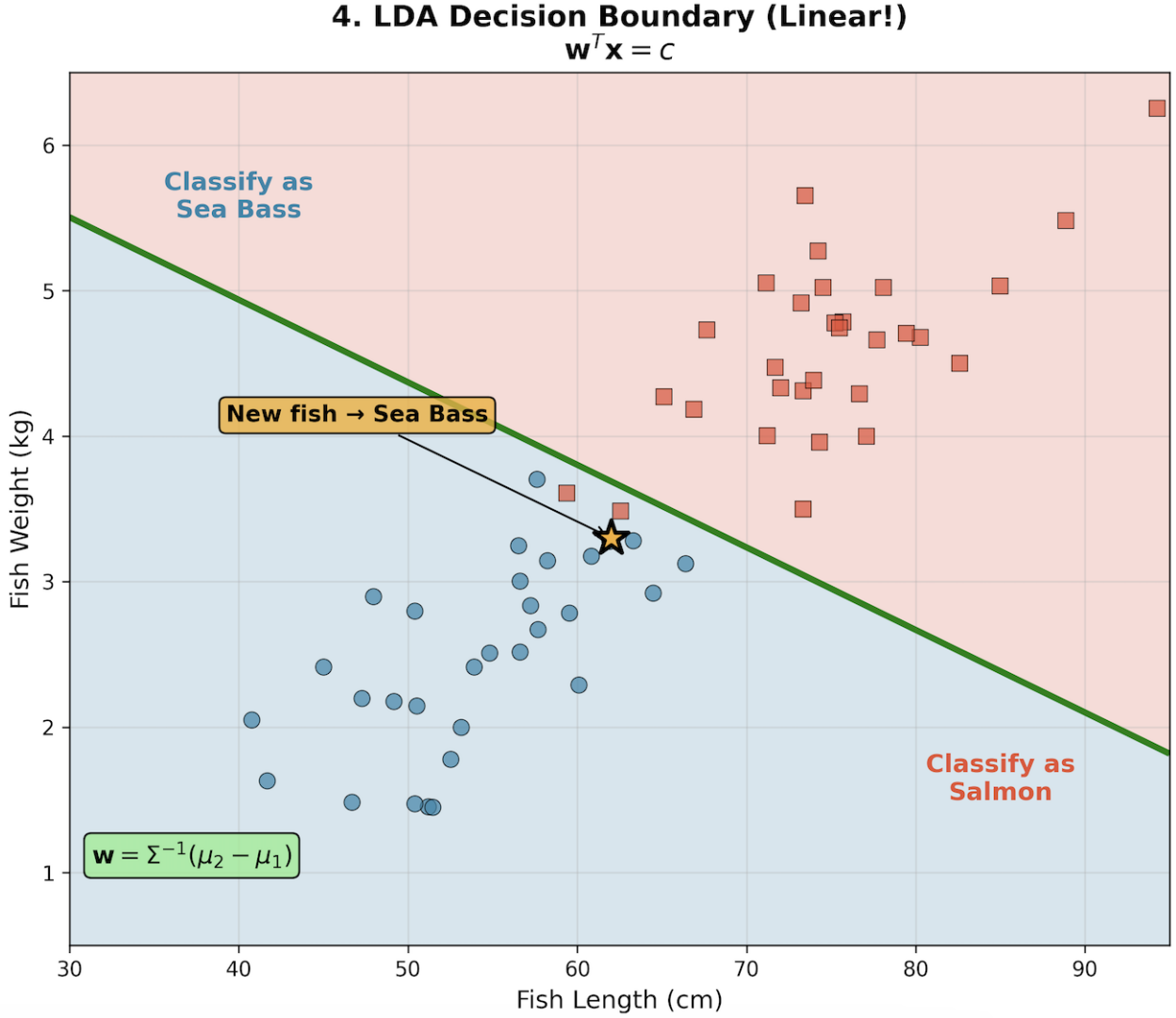
Linear Discriminant Analysis (2D case)





Linear Discriminant Analysis (2D case)



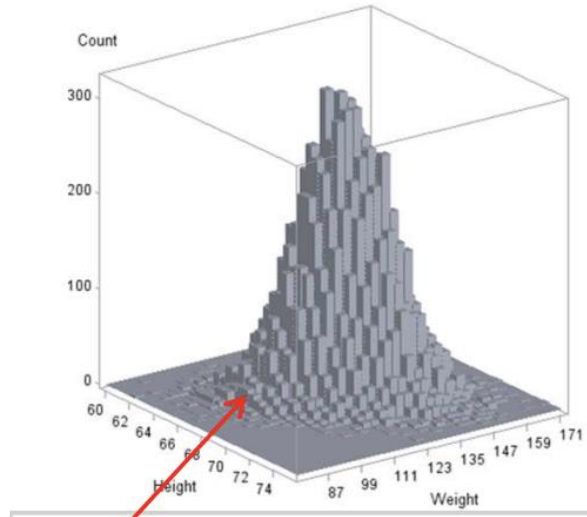


Multivariate Linear Discriminant Analysis (2D case)

$$f_i(\mathbf{x}) = \frac{1}{\sqrt{2\pi}^n \sqrt{\det \widehat{\boldsymbol{\Sigma}}}} \exp\left(-\frac{1}{2}(\mathbf{x} - \widehat{\boldsymbol{\mu}_i})^T \widehat{\boldsymbol{\Sigma}}^{-1}(\mathbf{x} - \widehat{\boldsymbol{\mu}_i})\right)$$

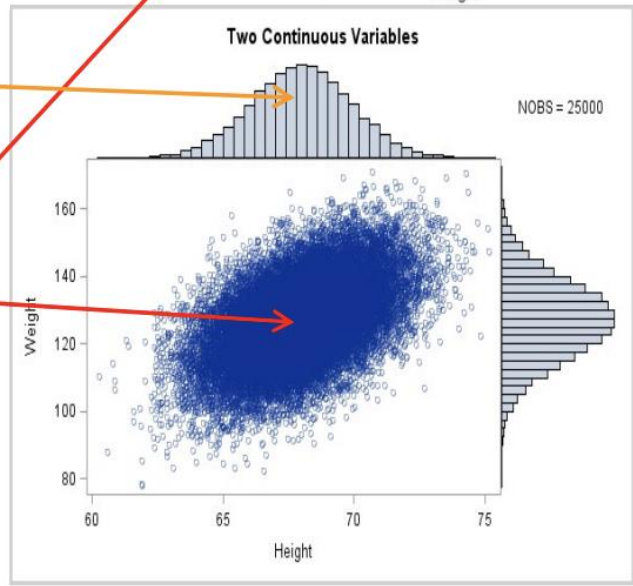
$$\widehat{\boldsymbol{\mu}}_i = \frac{1}{m}\sum_{j=1}^{m} \boldsymbol{x}_j \ , \quad \boldsymbol{x}_j \in \mathbb{R}^n$$

$$\hat{\Sigma}(a,b) = \frac{1}{m-1}\sum_{j=1}^{m}\left(x_{aj} - \hat{\mu}_a\right)\left(x_{bj} - \hat{\mu}_b\right)$$



Marginal distribution of height

Simultaneous distribution of 25000 corresponding values of height and weight



Estimation of parameters

We sample the following, where – in this case - each column is an observation and each row a variable:

$$X = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 4 & 5 \\ 1 & 5 & 9 \end{bmatrix}$$

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$$X = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 4 & 5 \\ 1 & 5 & 9 \end{bmatrix}$$

$$\mathbf{X}_i = \begin{bmatrix} X_{i1} \\ \vdots \\ X_{ip} \end{bmatrix}$$

$$\rho_{ij} = \operatorname{Corr}(X_i, X_j) = \frac{\operatorname{Cov}(X_i, X_j)}{\sqrt{\operatorname{V}(X_i)\operatorname{V}(X_j)}}.$$

$$\bar{\mathbf{X}} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{X}_i = \begin{bmatrix} \bar{X}_1 \\ \vdots \\ \bar{X}_p \end{bmatrix}$$

$$\mathbf{S} = \frac{1}{n-1}\sum_{i=1}^{n} (\mathbf{X}_i - \bar{\mathbf{X}})(\mathbf{X}_i - \bar{\mathbf{X}})^T = \frac{1}{n-1}\sum_{i=1}^{n} \mathbf{X}_i\mathbf{X}_i^T - \frac{n}{n-1}\bar{\mathbf{X}}\,\bar{\mathbf{X}}^T.$$

$$\hat{\Sigma}(a,b) = \frac{1}{m-1}\sum_{j=1}^{m} \left(x_{aj} - \hat{\mu}_a\right)\left(x_{bj} - \hat{\mu}_b\right)$$

The mean

$$X = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 4 & 5 \\ 1 & 5 & 9 \end{bmatrix}$$

$$\begin{bmatrix} 3 \\ 12 \\ 15 \end{bmatrix}$$

$$\begin{bmatrix} 1.5 \\ 6 \\ 7.5 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} \frac{5}{3} & 3 & 5\frac{1}{3} \end{bmatrix}$$

The covariance matrix

$$X = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 4 & 5 \\ 1 & 5 & 9 \end{bmatrix}$$

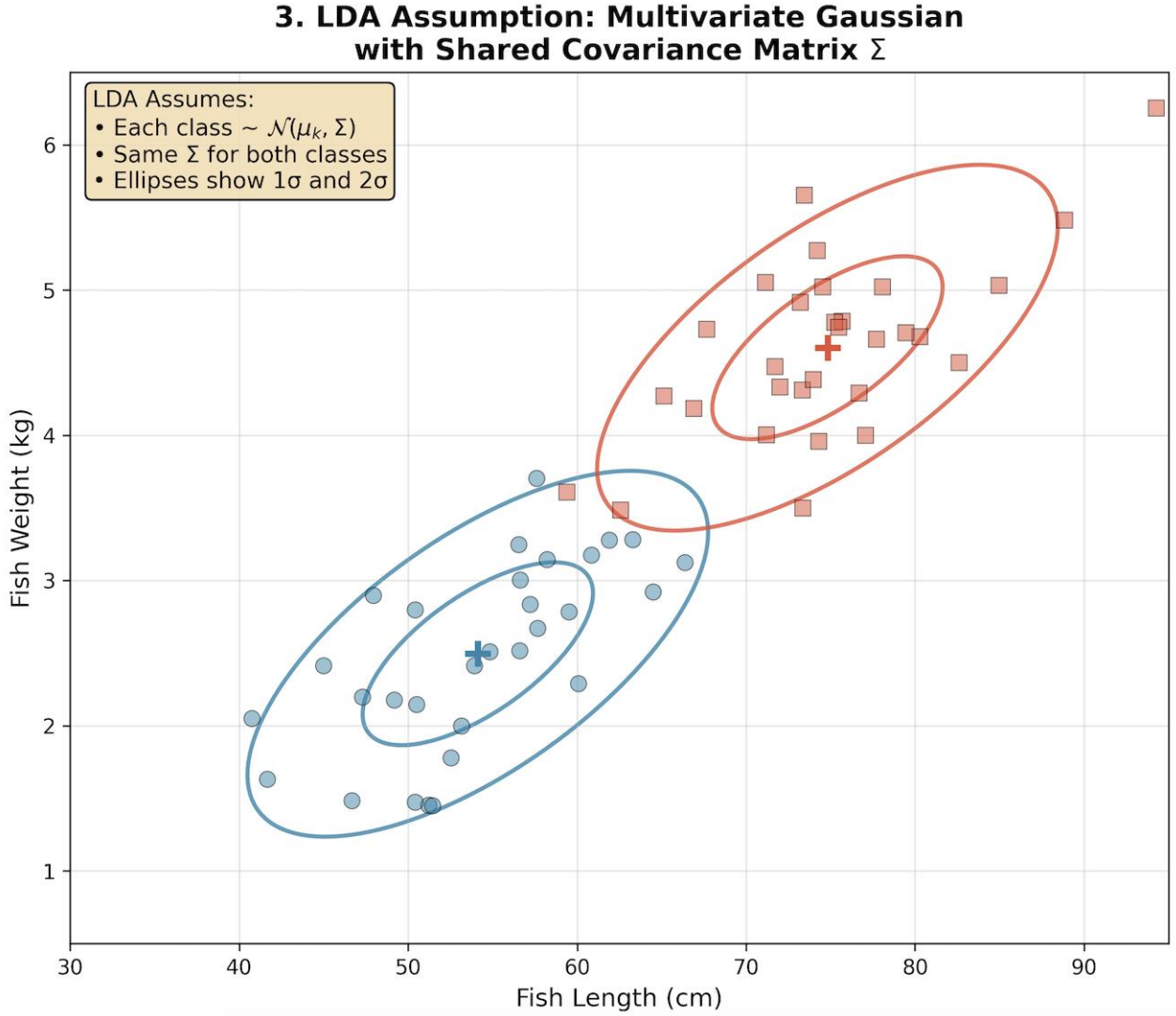
$$\begin{bmatrix} 2 & 1 & 4 \\ 1 & 2 & 8 \\ 4 & 8 & 32 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0.5 & 0.5 \\ 0.5 & 1 & 1 \\ 0.5 & 1 & 1 \end{bmatrix}$$

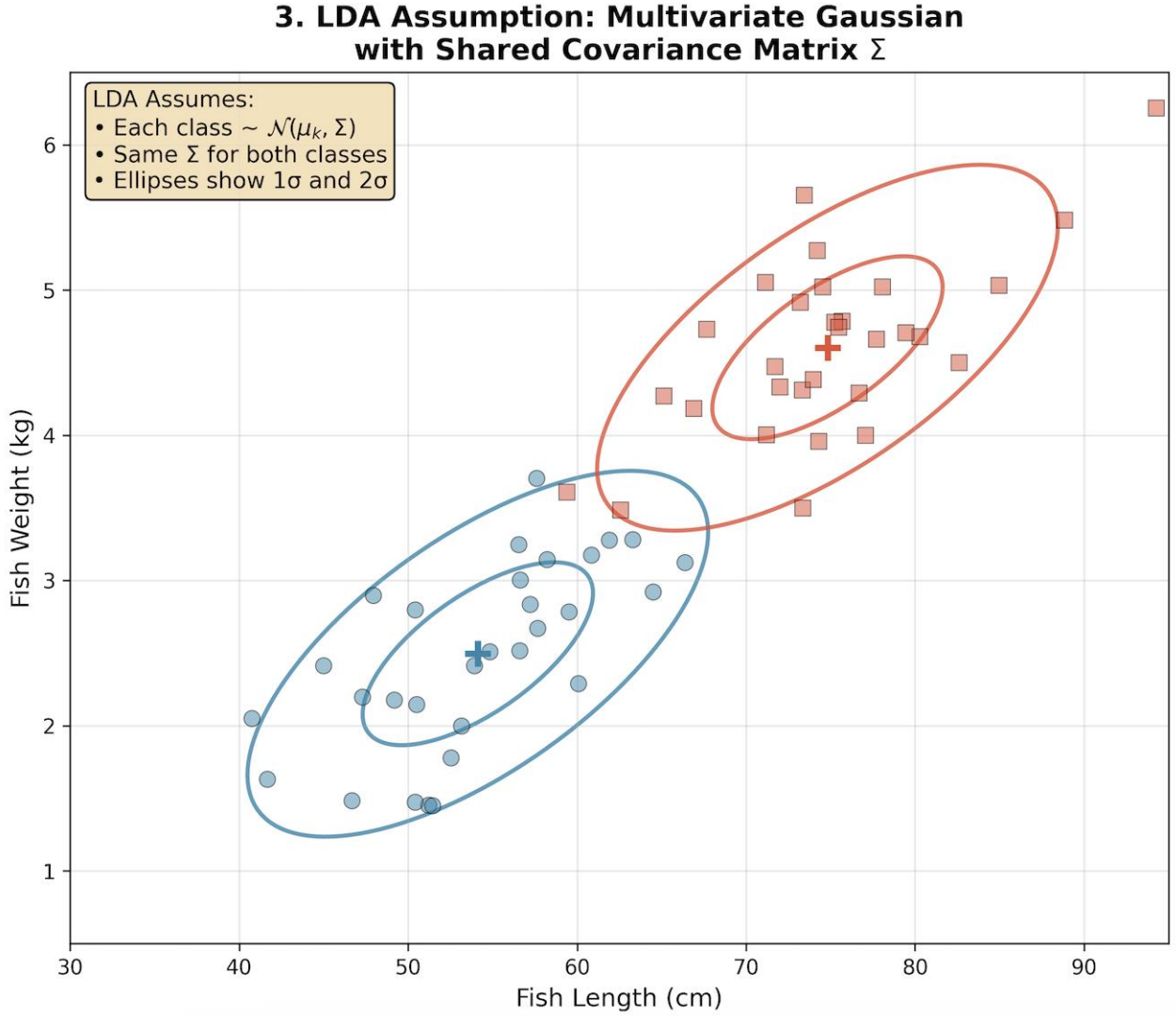
$$\begin{bmatrix} 1 & 0.5 & 2 \\ 0.5 & 1 & 4 \\ 2 & 4 & 16 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 1 \\ 16 \end{bmatrix}$$

Linear Discriminant Analysis (2D case)



Linear Discriminant Analysis (2D case)



Class mean vectors (MLE):

$$\hat{\boldsymbol{\mu}}_1 = \frac{1}{n_1}\sum_{i=1}^{n_1}\mathbf{x}_i^{(1)} = [54.12,\ 2.50]^T \quad \text{(Sea Bass)}$$

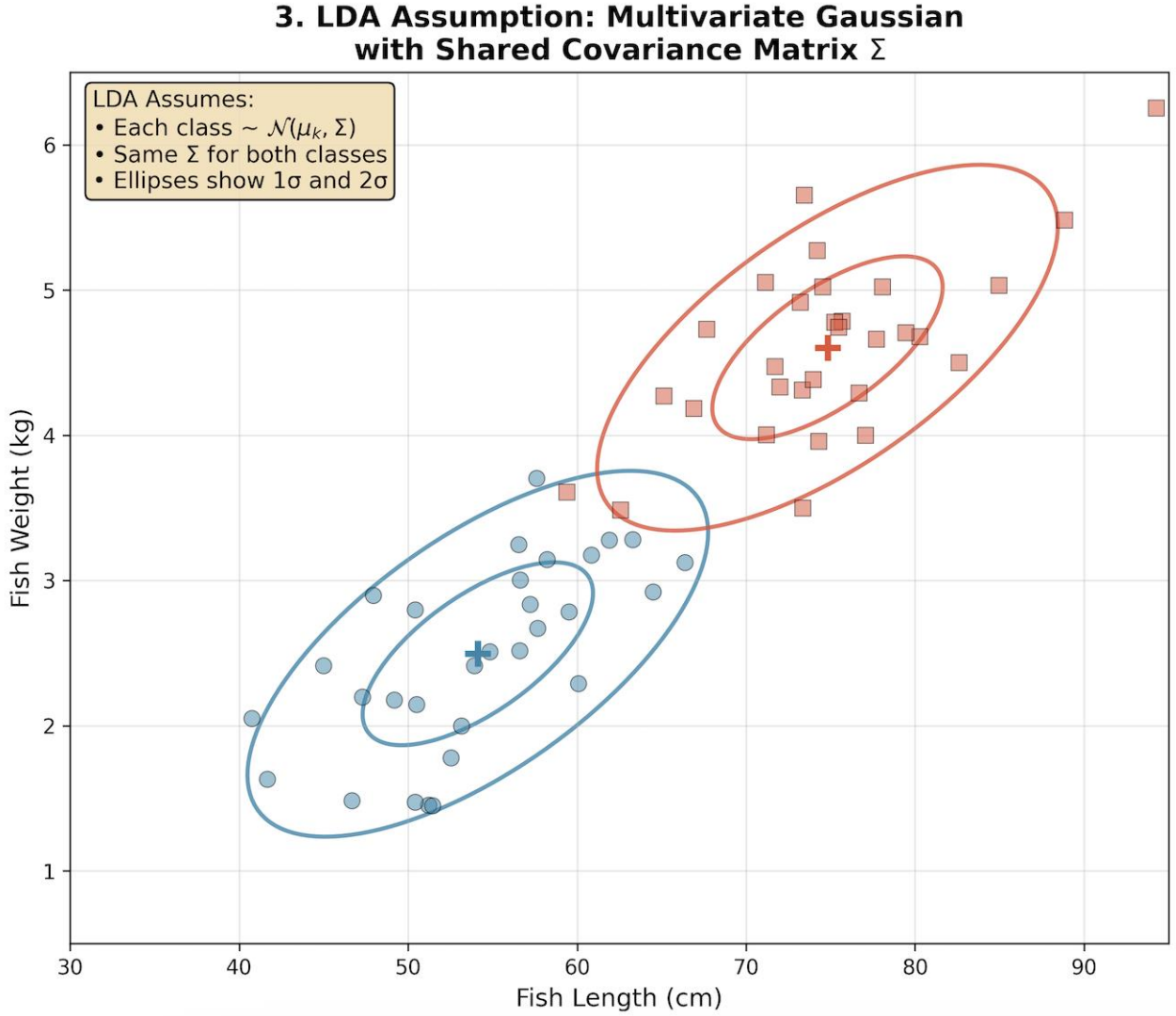
$$\hat{\boldsymbol{\mu}}_2 = \frac{1}{n_2}\sum_{i=1}^{n_2}\mathbf{x}_i^{(2)} = [74.82,\ 4.60]^T \quad \text{(Salmon)}$$

Pooled covariance matrix estimate:

$$\hat{\Sigma} = \frac{(n_1-1)\mathbf{S}_1 + (n_2-1)\mathbf{S}_2}{n_1+n_2-2}$$

where $\mathbf{S}_k = \frac{1}{n_k-1}\sum_{i=1}^{n_k}(\mathbf{x}_i^{(k)} - \hat{\boldsymbol{\mu}}_k)(\mathbf{x}_i^{(k)} - \hat{\boldsymbol{\mu}}_k)^T$ is the sample covariance

Linear Discriminant Analysis (2D case)

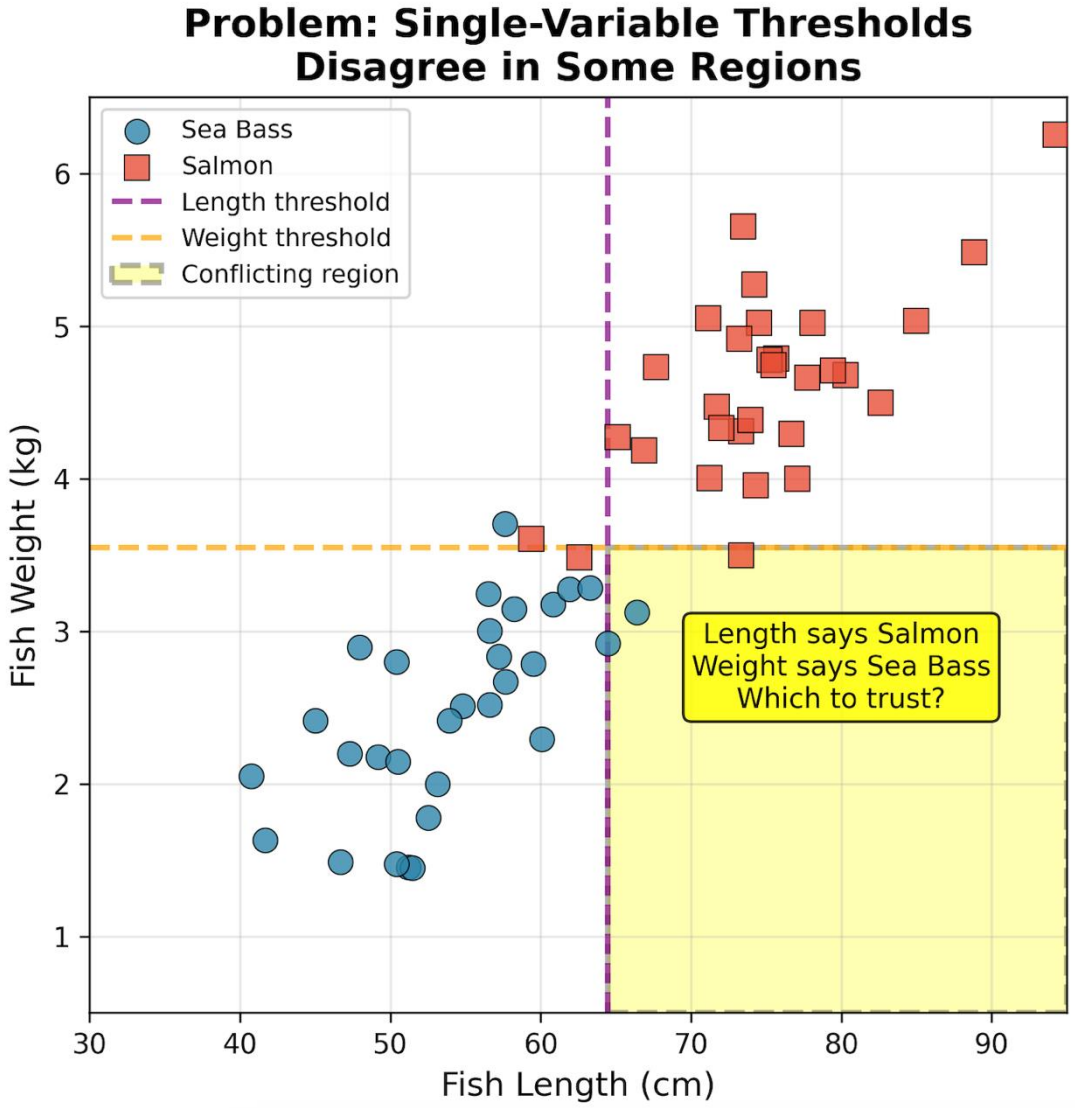


Classification strategies

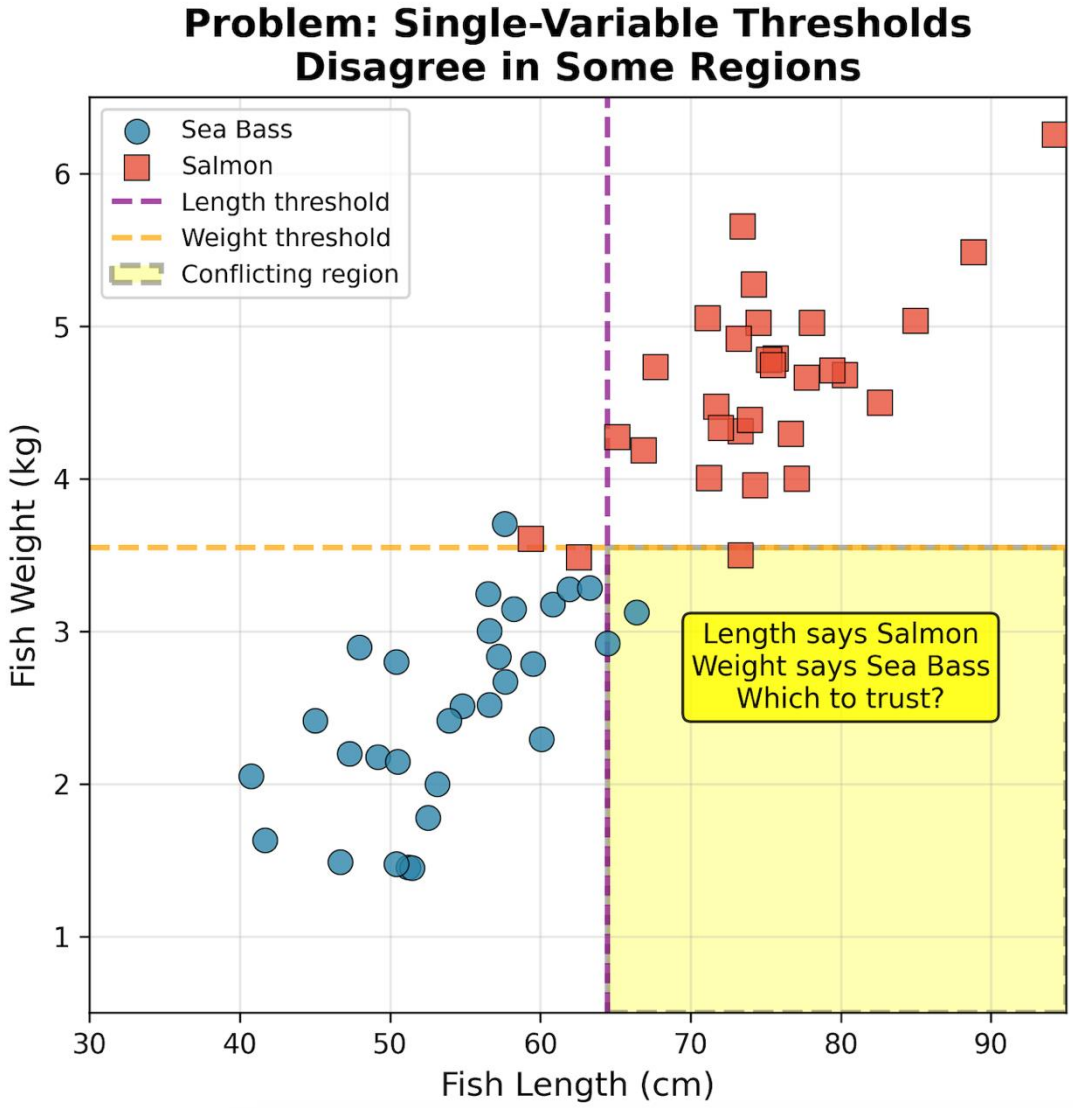
- Simple threshold

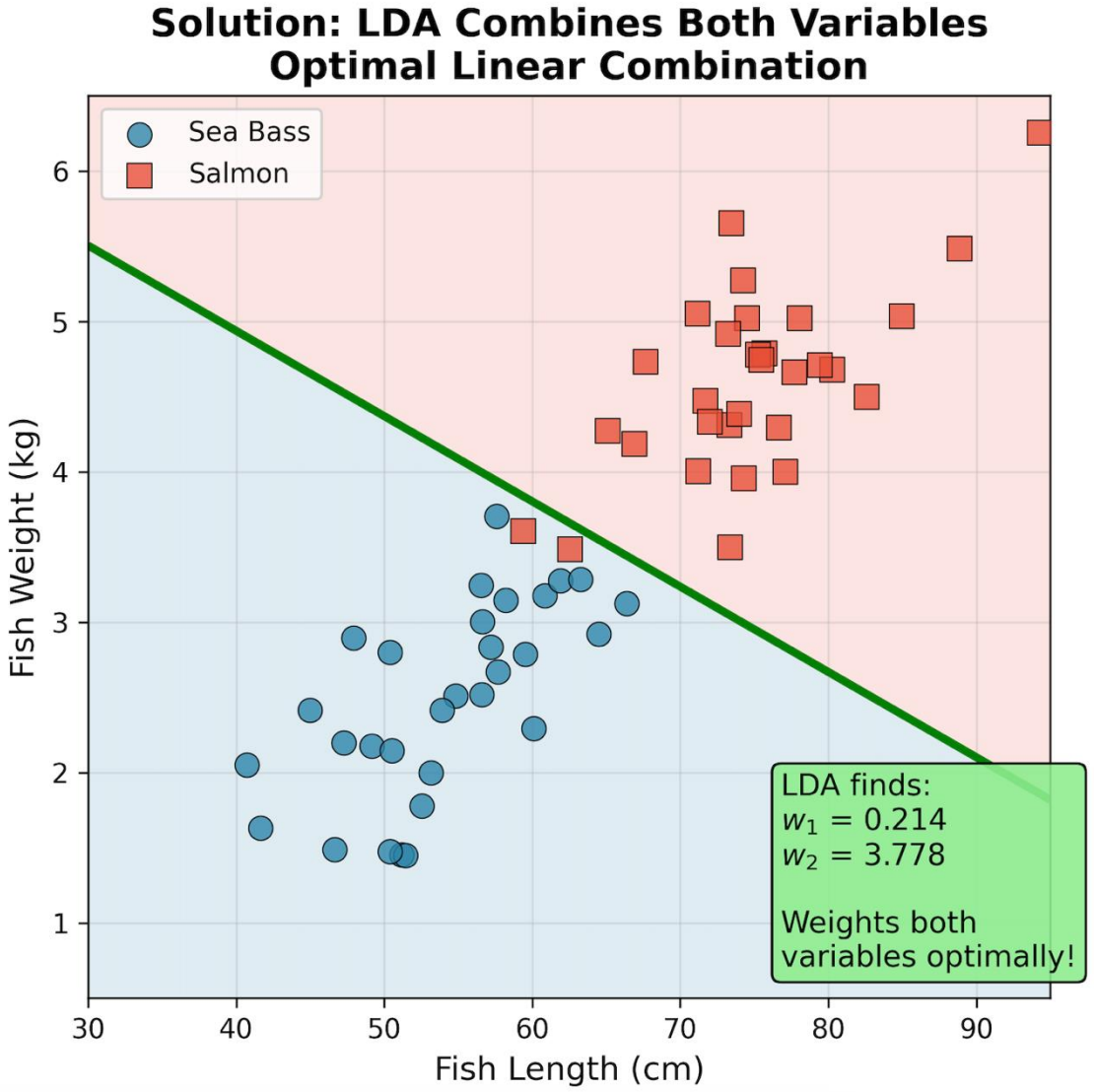
- Multiple thresholds??

Linear Discriminant Analysis (2D case)

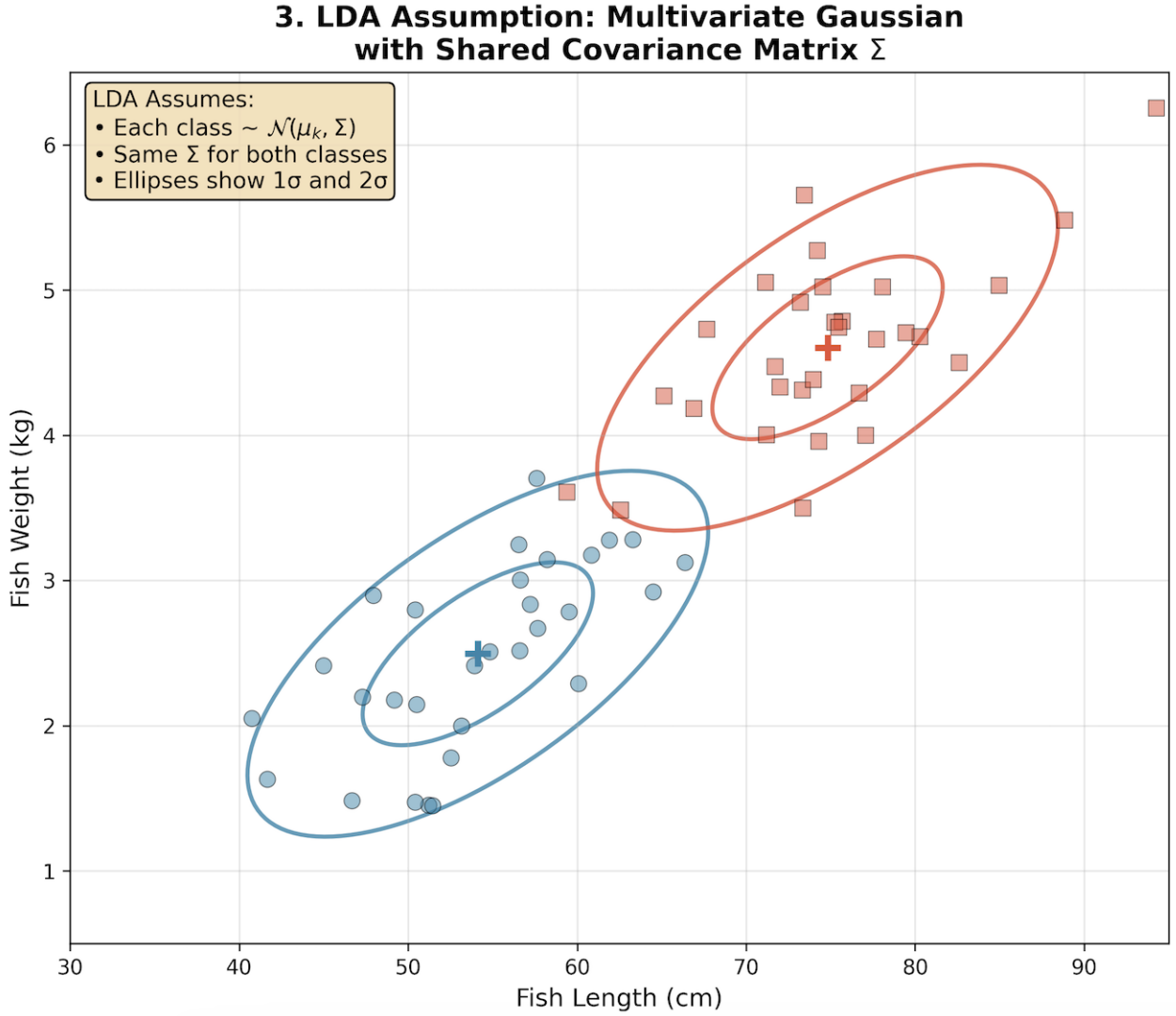


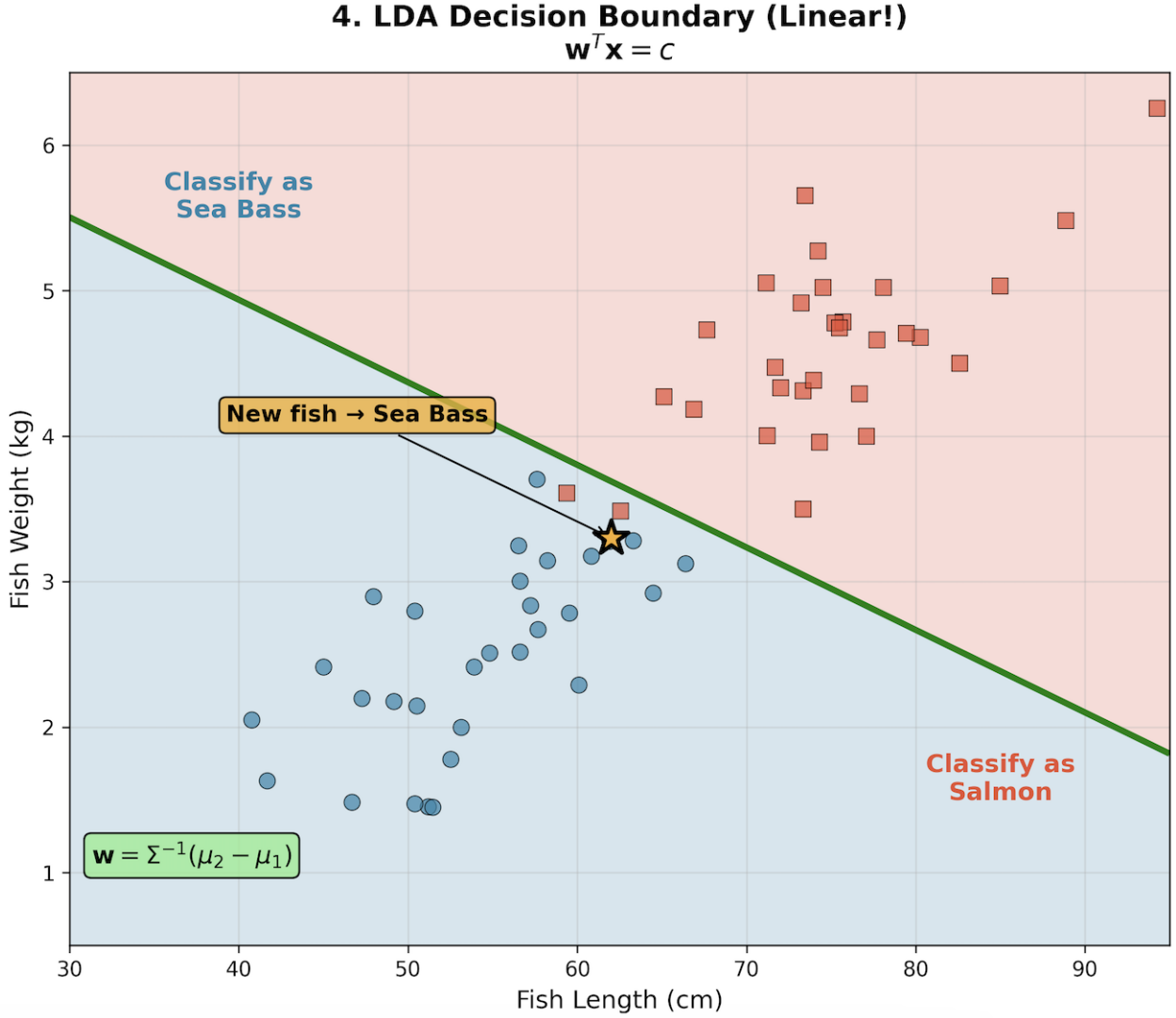
Linear Discriminant Analysis (2D case)



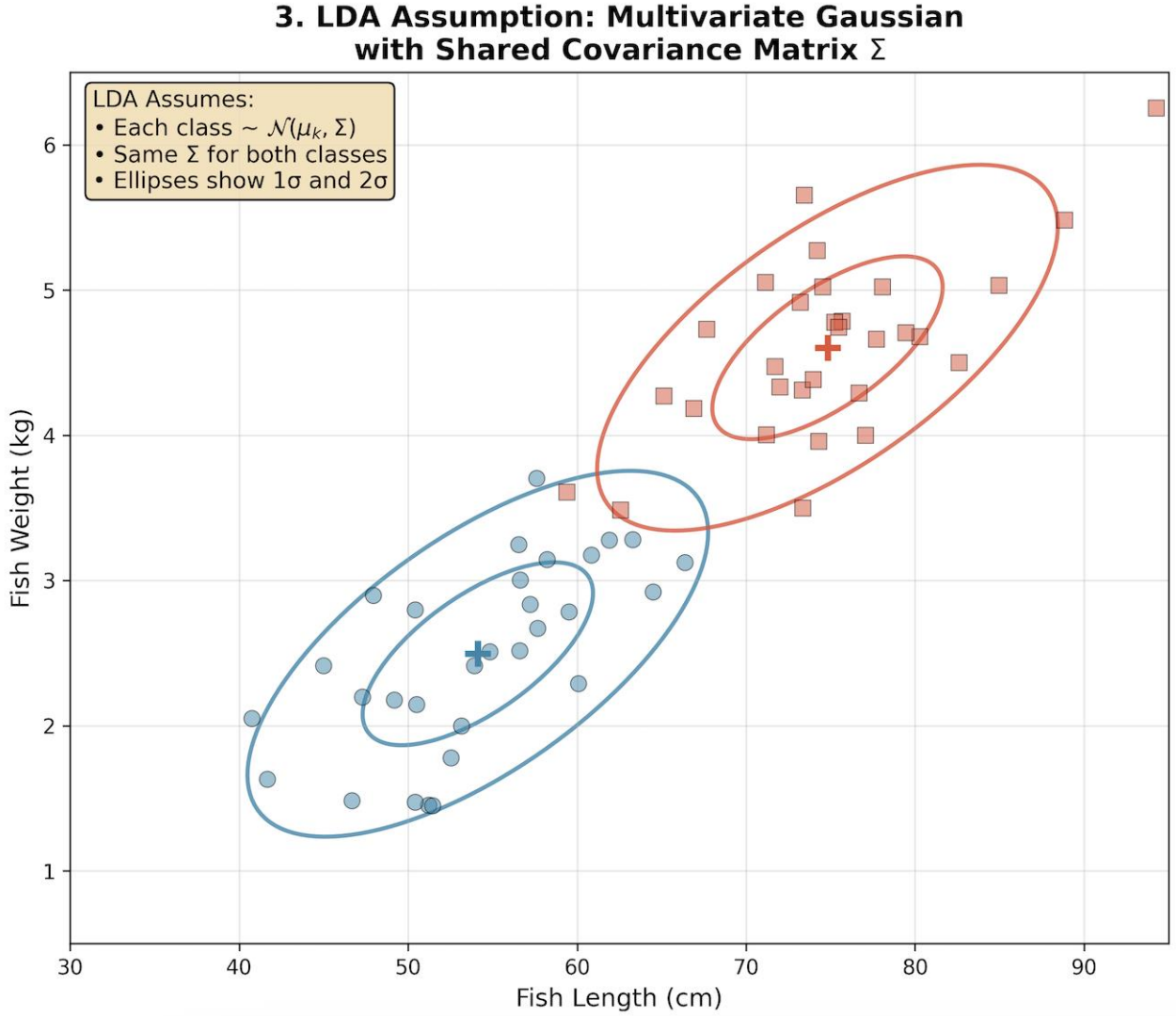


Linear Discriminant Analysis (2D case)





Linear Discriminant Analysis (2D case)



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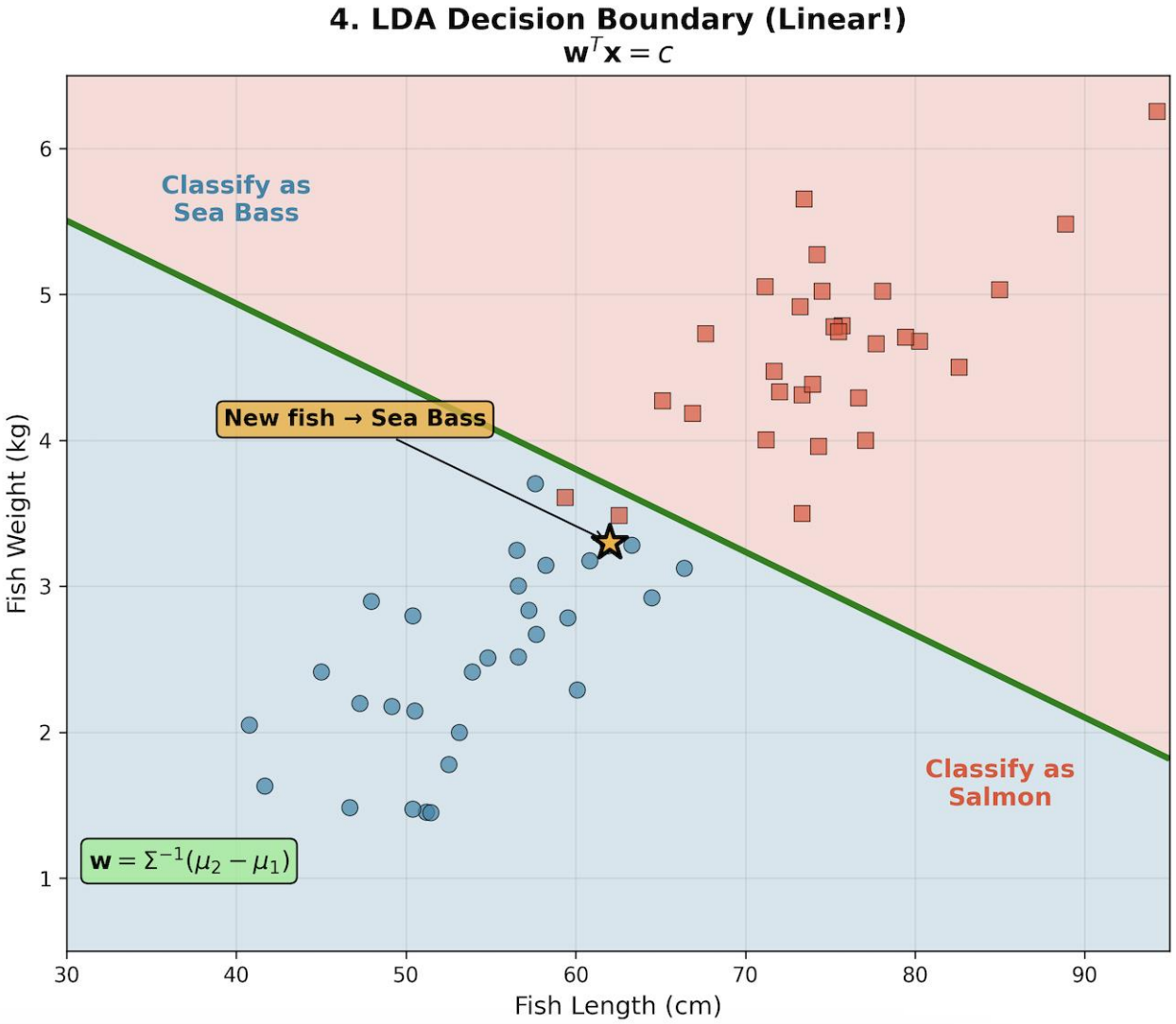
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Pooled covariance matrix estimate:

$$\hat{\Sigma} = \frac{(n_1-1)\mathbf{S}_1 + (n_2-1)\mathbf{S}_2}{n_1+n_2-2}$$

where $\mathbf{S}_k = \frac{1}{n_k-1}\sum_{i=1}^{n_k}(\mathbf{x}_i^{(k)} - \hat{\boldsymbol{\mu}}_k)(\mathbf{x}_i^{(k)} - \hat{\boldsymbol{\mu}}_k)^T$ is the sample covariance

Linear Discriminant Analysis (2D case)



Taking the log of the posterior ratio and simplifying:

$$\log \frac{P(\text{class } 2 \mid \mathbf{x})}{P(\text{class } 1 \mid \mathbf{x})} = \log \frac{\pi_2}{\pi_1} + (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^T \Sigma^{-1} \mathbf{x} - \frac{1}{2}(\boldsymbol{\mu}_2^T \Sigma^{-1} \boldsymbol{\mu}_2 - \boldsymbol{\mu}_1^T \Sigma^{-1} \boldsymbol{\mu}_1)$$

This simplifies to a linear function of $\mathbf{x}$:

$$\delta(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + w_0$$

where:

$$\mathbf{w} = \Sigma^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1) \quad \text{and} \quad w_0 = \log \frac{\pi_2}{\pi_1} - \frac{1}{2}(\boldsymbol{\mu}_2 + \boldsymbol{\mu}_1)^T \mathbf{w}$$

Classification - Evaluation

1. **Resubstitution** classification of the training data set: Each observation in the training data set is classified using the estimated discriminant function.
   – Too optimistic results – the model has already seen the data

2. **Cross Validation** of the training data set: Each observation in the training data set is classified using a discriminant function computed from the other observations in the training data set excluding the observation being classified.
– A good alternative when data is scarce

3. **Set Aside** classification: Divide the input data set randomly into two data sets, the training and the test data set. Estimate the discriminant functions on the training data set and test them on the test data set.
- the gold standard, but often not possible due to lack of data

Classification - Evaluation

Model selection: estimating the performance of different models in order to choose the best one.

Model assessment: having chosen a final model, estimating its prediction error (generalization error) on new data.



Classification - Evaluation

Number of observations from		Classified as							Sum
		π_1	$\cdots$	π_i	$\cdots$	π_j	$\cdots$	π_k	
Nature	π_1	N_{11}	$\cdots$	N_{1i}	$\cdots$	N_{1j}	$\cdots$	N_{1k}	$N_{1\cdot}$
	$\vdots$	$\vdots$		$\vdots$		$\vdots$		$\vdots$	$\vdots$
	π_i	N_{i1}	$\cdots$	N_{ii}	$\cdots$	N_{ij}	$\cdots$	N_{ik}	$N_{i\cdot}$
	$\vdots$	$\vdots$		$\vdots$		$\vdots$		$\vdots$	$\vdots$
	π_j	N_{j1}	$\cdots$	N_{ji}	$\cdots$	N_{jj}	$\cdots$	N_{jk}	$N_{j\cdot}$
	$\vdots$	$\vdots$		$\vdots$		$\vdots$		$\vdots$	$\vdots$
	π_k	N_{k1}	$\cdots$	N_{ki}	$\cdots$	N_{kj}	$\cdots$	N_{kk}	$N_{k\cdot}$
Sum		$N_{\cdot 1}$	$\cdots$	$N_{\cdot i}$	$\cdots$	$N_{\cdot j}$	$\cdots$	$N_{\cdot k}$	$N_{\cdot\cdot}$

Table 5.2 – Confusion matrix showing the result of classifying $N_{\cdot\cdot}$ observations. N_{ij} is the number of observations from population π_i classified as coming from population π_j.

Measure	Formula	Probabilistic interpretation
Global accuracy (ACC)	$\frac{1}{N_{\cdot\cdot}} \sum_{i=1}^{k} N_{ii}$	Probability of correct classification
Global error rate, misclassification rate (MIS)	$1 - \frac{1}{N_{\cdot\cdot}} \sum_{i=1}^{k} N_{ii} = \frac{1}{N_{\cdot\cdot}} \sum_{i \neq j} N_{ij}$	Probability of misclassification
Class accuracy	$\frac{N_{ii}}{N_{i\cdot}}$	Conditional probability of being classified as π_i given true class is π_i.
Class error rate, misclassification rate	$1 - \frac{N_{ii}}{N_{i\cdot}} = \frac{N_{i\cdot} - N_{ii}}{N_{i\cdot}}$	Conditional probability of not being classified as π_i given true class is π_i

Table 5.3 – Error rates estimated from the confusion matrix

Experiments

- Implement the mathematical model – linear discriminant analysis
- Find the model parameters – day 1 is the training set
- Classify pixels in the training data – test if the model works at all
- Classify pixels in the entire salami for day 1
- Classify data for the remaining days
- Find the best day to train on
- Impact of prior knowledge

Analysis

- Quantitative analysis with regards to manual annotations
- Qualitative analysis by looking at the segmentation
- Consider - among others - the following
  - How good should the results be?
  - What are we comparing to?
  - How good is the model?
  - What could improve the measurement of meat and fat?

Report

Coherent report which independently describes the analysis

- Introduction with problem statement
- Description of method – the mathematical model (general)
- Description of experiment, analysis, and results (concrete)
- Discussion of results:
  - Does the method solve the problem?
  - How good is the solution?
  - What parameters have been used?
  - Are the data representative of the problem?
  - What are the limitations of the method?
- Conclusion – what is the contribution of the analysis

Questions??

Scientific Writing - Overview

1. Why is it important?
2. Organization - Structure
3. Resources
4. Techniques and Examples
5. The fine line between references, citations, and plagiarizing

Scientific Writing - Overview

1. **Why is it important?**
2. Organization - Structure
3. Resources
4. Techniques and Examples
5. The fine line between references, citations, and plagiarizing

You'll Never Stop Writing


24

4. How to Set Up a MoM/MLFMM Analysis

4.1 MoM/MLFMM Object Setup

When setting up a MoM/MLFMM object in GRASP only two attributes need to be set by the user: FREQUENCY and SCATTERER. All other attributes are pre-specified and can be left as they are to begin with. Their default values are assessed to yield a reasonable compromise between accuracy and speed, and their meaning will be discussed in the following.

A MoM/MLFMM object with FREQUENCY and SCATTERER specified by the user, as shown in Figure 4-1, is thus ready to be used as the Target in a Get Currents command without further modifications.

Figure 4-1 MoM/MLFMM object with FREQUENCY and SCATTERER attributes set by the user.

Software documentation



Design of Axially Corrugated Horn for Ka-Band Offset Reflector Antenna

Technical Note 1

Provided by: TICRA, Denmark

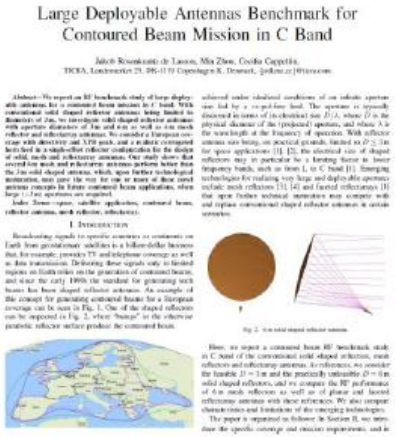
Design/Consulting reports



4. Feed array optimization and radiometer characteristics (WP5300, TICRA)

4.1 Review of radiometer design and analysis approaches

R&D project proposals and reports



Large Deployable Antennas Benchmark for Contoured Beam Mission in C Band

Conference papers

(...Some of it even in LaTeX 😮 *)*



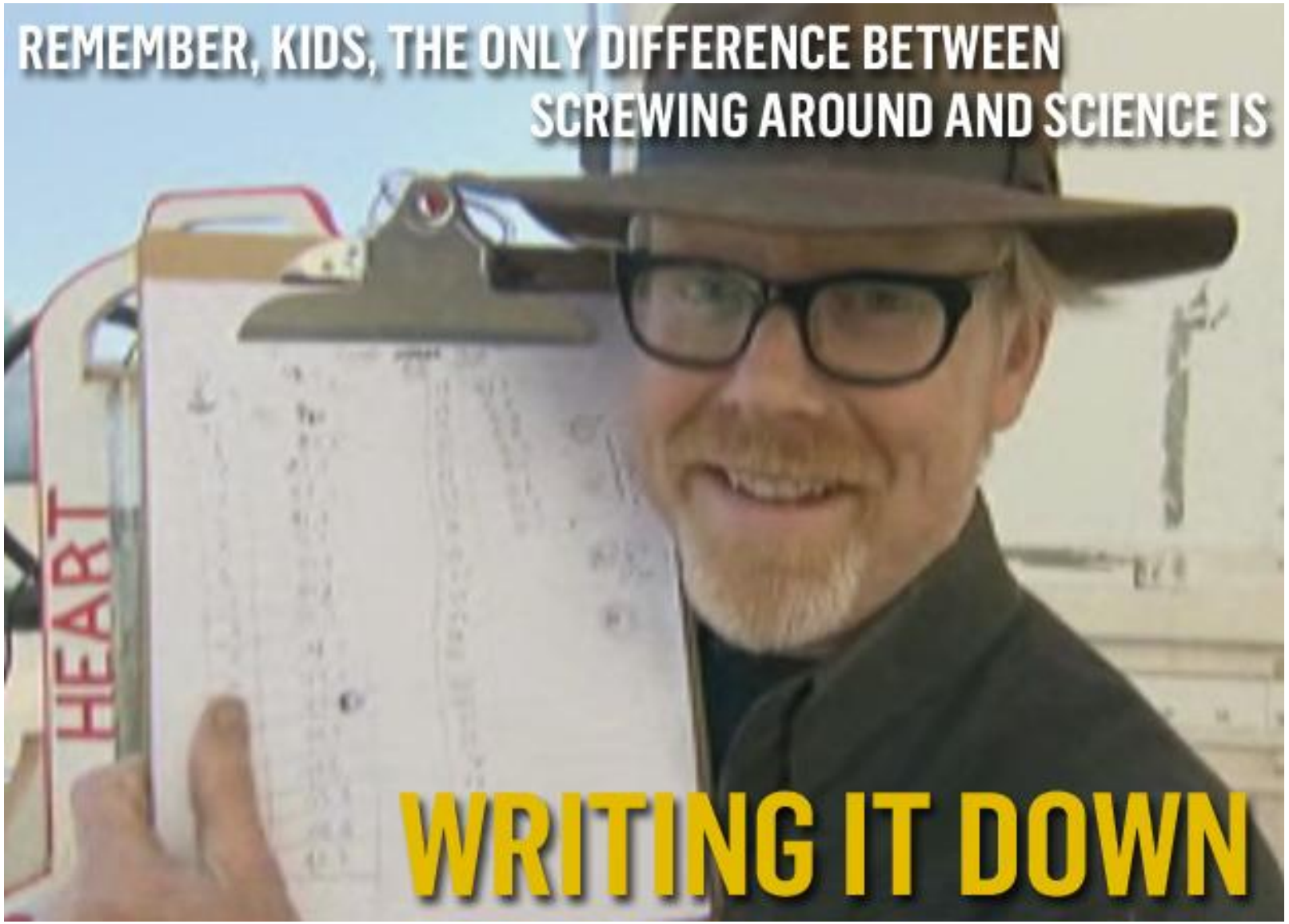
Key Skill: Concise Writing

Scientific Writing

- **Scientific writing** is not just writing about science; it is the technical writing that scientists do to communicate their research to others

- Analysis and mathematical models are useless...
  – when they are not accessible for others – or almost useless

- They get value when they can be used by others!



[Adam Savage, Mythbusters]

Scientific Writing - Overview

1. Why is it important?
2. **Organization – Structure**
3. Techniques and Examples
4. Resources
5. The fine line between references, citations, and plagiarizing

Treat your reader as you would a guest in your house

Anticipate their needs: would you like something to drink?

Something to eat? Perhaps now, after eating you would like to rest?



[Slides from Bill Freeman, MIT,
How to write a good CVPR submission]

Structure

- Organization of your writing is critical

- Why?
  - If the structure is as anticipated
    - Makes the report (article) easier to read
    - Makes the report (article) easier to write

- Based on recommandations for scientific articles [1] and Global Denmark A/S
  - Must be understandable by professionals/researchers within the same field
  - Must be precise enough to allow others to reproduce the results
  - Must be fast to read
  - Must be as understandable as possible

Structure

- Introduction
- Methods and Materials
- Experiments, Results and Discussion
- Conclusion

Structure

- Abstract
- Introduction
- Related work
- Methods and Materials
- Experiments and Results
- Discussion
- Conclusion

Structure

- **Abstract**
- Introduction
- Related work
- Methods and Materials
- Experiments and Results
- Discussion
- Conclusion

Abstract

We trained a large, deep convolutional neural network to classify the 1.2 million high-resolution images in the ImageNet LSVRC-2010 contest into the 1000 different classes. On the test data, we achieved top-1 and top-5 error rates of 37.5% and 17.0% which is considerably better than the previous state-of-the-art. The neural network, which has 60 million parameters and 650,000 neurons, consists of five convolutional layers, some of which are followed by max-pooling layers, and three fully-connected layers with a final 1000-way softmax. To make training faster, we used non-saturating neurons and a very efficient GPU implementation of the convolution operation. To reduce overfitting in the fully-connected layers we employed a recently-developed regularization method called "dropout" that proved to be very effective. We also entered a variant of this model in the ILSVRC-2012 competition and achieved a winning top-5 test error rate of 15.3%, compared to 26.2% achieved by the second-best entry.

- Typically 150-300 words
- Purposes for the readers:
  - decide to read the full article
  - prepare them for the details of the full paper
- Standalone
- Omitted in short articles/reports

[Abstract from Krizhevsky et al, "ImageNet Classification with Deep Convolutional Neural Networks" NeurIPS 12]

Structure

- Abstract
- **Introduction**
- Related work
- Methods and Materials
- Experiments and Results
- Discussion
- Conclusion

Structure

- Abstract
- **Introduction**
- Related work
- Methods and Materials
- Experiments and Results
- Discussion
- Conclusion

Introduction:

- ***Motivation for the work***
- ***Prepares readers for the structure***

Introduction: 4 components

probably (not necessarily) 4 paragraphs

1. **Context**
2. **Need**
3. **Task**
4. **Object**

An effective introduction for a paper by Nature, Scitable

Context		With approximately 243 million cases and 863 000 attributed deaths reported globally in 2009, malaria is one of the most severe infectious diseases, primarily affecting the world's most disadvantaged populations. Of the four typically recognized *Plasmodium* species causing disease in humans, *Plasmodium falciparum* causes most mortality, mainly in children below the age of 5, and *Plasmodium vivax* most morbidity, additionally representing a reservoir of latent infection that hampers current control and future elimination efforts.
Need	*what we have*	No new class of antimalarials has been introduced into clinical practice since 1996, because of the intrinsic difficulties in discovering and developing new antimicrobials, as well as a relative lack of public and private resource commitment towards antimalarial research. Today, the last class of widely efficacious drugs, the artemisinins, is being compromised by the rise of *P. falciparum* strains with reduced clinical response to artemisinin-containing drug combinations. The genomics revolution has not yet led to new antimalarial medicines and target-based lead discovery has produced disappointing results, generally for lack of whole-cell activity as documented for antibacterials.
	what we want	To secure this property in all chemical starting points for new antimalarial leads,
Task		we have tested the approximately two-million-compound library used for high-throughput screening at GlaxoSmithKline (GSK) for inhibitors of *P. falciparum*'s intraerythrocytic cycle, the parasite's growth phase responsible for disease symptoms, which is amenable to *in vitro* culture.
Object of the document		This paper describes 13 533 compounds confirmed to inhibit parasite growth by more than 80% at 2 mM concentration, 82% of which were proprietary and thus unknown to the general research community.

Introduction: 4 components

probably (not necessarily) 4 paragraphs

1. **Context** (provide context to readers, establish importance of work)
2. **Need** (state the need of the work, what the community currently has and what it wants)
3. **Task** (what you have done to address the need)
4. **Object** (preview the remainder of the report to mentally prepare readers)

Introduction

- Describes the problem – why bother with the investigation/modelling
- Describes relevance – who benefits from work (ask why?)
  - *"We wish to suggest a structure for the salt of deoxyribose nucleic acid (D.N.A.). This structure has novel features which are of considerable biological interest." Crick & Watson, Nature, 1953*
- Describes related work – how has others approached the task (ask how?)
- Argues for the choice of method

Structure

- Abstract
- Introduction
- **Related work**
- Methods and Materials
- Experiments and Results
- Discussion
- Conclusion

Structure

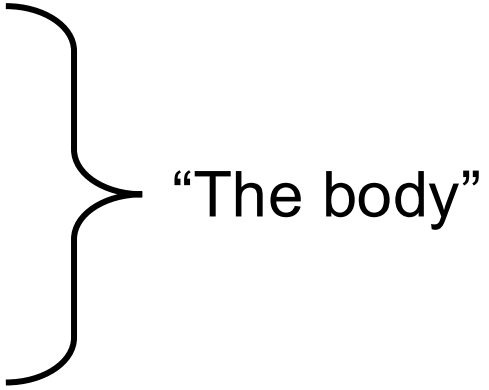
- Abstract
- Introduction
- **Related work**
- Methods and Materials
- Experiments and Results
- Discussion
- Conclusion

- Often in scientific papers, long articles, or thesis
- Description of the related work in details
- How our work contributes something new

- **Do not** list sentences that summarize a single paper
- **Summaries:** useful way to build a picture for yourself of existing related work

Structure

- Abstract
- Introduction
- Related work
- **Methods and Materials**
- Experiments and Results
- Discussion
- Conclusion



Methods and materials

- Describes the method/model in details (the mathematics are described)
- The description should be detailed enough for others to **reproduce** the experiments
- **Reference** to known methods – no need to describe everything
- Results should ***not*** be described here
- The method description is **general**
  - Materials are **not**

Methods and materials

- Method sections are boring to read, descriptive
  - yet they need not be

- Make this part interesting:
  - explain your choices
  - what is special or different in your approach, why?
  - move detailed, less important parts to the end of the report (Appendix)

Structure

- Abstract
- Introduction
- Related work
- Methods and Materials
- **Experiments and Results**
- Discussion
- Conclusion

Experiments and Results

- Describes data, experiments and results
- Others should be able to reproduce the experiments
- Data and results are described concise
  - Use statistics – mean values, standard deviations, confidence intervals etc.
  - Use representative examples
  - Point out general trends
  - Point out deviations, surprises and failure cases
- Method should ***not*** be described here
- Experiments and results are specific

Structure

- Abstract
- Introduction
- Related work
- Methods and Materials
- Experiments and Results
- **Discussion**
- Conclusion

Discussion

- Principles, relations, and general conditions shown in results
- Stress exceptions
- Relate to what was expected
- Discuss usefulness of the analysis/model
- (Future work / extensions)

Results and ***Discussion*** are often combined because results make little sense to most readers without interpretation.

Structure

- Abstract
- Introduction
- Related work
- Methods and Materials
- Experiments and Results
- Discussion
- **Conclusion**

Conclusion

- Most important outcome of your work
- What are the main results
- Contribution with concise argumentation: What do we know now, that we did not know before

- **Not a summary**
- **Do not** restate what you have done or what the paper does.
- Instead, focus on what you have **found** and, especially, on what your **findings mean**

Scientific Writing - Overview

1. Why is it important?
2. Organization – Structure
3. **Techniques and Examples**
4. Resources
5. The fine line between references, citations, and plagiarizing

Techniques

- Language – not literature, but must be easily readable
- Avoid long sentences with a lot of conjunctions
- Break up long blocks of text into simpler ones:
  - Few lines and verbs per sentence; few sentences per paragraph.
  - **2-3-4 rule:** Consider splitting every sentence of more than 2 lines, every sentence with more than 3 verbs, and every paragraph with more than 4 "long" sentences.
  - Sentences of max 25 words (Good rule of thumb)

Techniques

- Choose the **active voice** more often than the **passive voice**:
  - An **analysis** was performed on the data
  - We **analyzed** the data
- Reduce **dead weight words** and phrases
- Hunt down and cast out **all unneeded words** that might slow your reader
- Use **abbreviations** correctly (e.g., cf., i.e., etc.)
- Use handles (Sec. 1, Fig. 4, Tab. 3)
- Follow: subject+verb+object (**SVO**)
- Use **strong verbs** and avoid turning verbs into nouns
- Eliminate **negatives**; use **positive** constructions instead

Techniques

- **Clarity** is key. Be **precise** and **consistent**!
  - "our method is much faster" is very vague.
  - It is better to quantify the improvement (e.g., "an order of magnitude faster").

- Repetitions are **GOOD**!
  - help the reader and remove ambiguities
  - more readable and easier to understand
  - do not use synonyms, always use the same word
    (e.g. If the wording you use at the beginning is "detecting recurrent motion patterns", consistently use it throughout the paper. Always use "recurrent", and not synonyms like "repetitive" or "frequent".)
    (e.g. Always use "motion", and not, say, "movement". If you switch from motion to movement, the reader might think you are actually talking about something different!)

Techniques

- Choose the **active voice** more often than the **passive voice**:
  - An **analysis** was performed on the data
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- Use **strong verbs** and avoid turning verbs into nouns
- Eliminate **negatives**; use **positive** constructions instead

Examples

Example of a bad sentence The main contributions of our method, which allow us to improve over the state-of-the-art, is the use of feature XXX, which is better than feature KKK, which was designed for synthetic data, but fails on real images, as it is not robust to clutter, and a new inference method based on ZZZ, which is much faster than any existing method, as shown by our experiments.

Examples

Example of a bad sentence The main contributions of our method, which allow us to improve over the state-of-the-art, is the use of feature XXX, which is better than feature KKK, which was designed for synthetic data, but fails on real images, as it is not robust to clutter, and a new inference method based on ZZZ, which is much faster than any existing method, as shown by our experiments.

- **Veryyyyyy long sentence**
- **no references**
- **no handles**
- **vague phrases (“much faster”, ”any existing method”, ”fails”)**
- **negatives: it is not robust**

Examples

Example of a bad sentence The main contributions of our method, which allow us to improve over the state-of-the-art, is the use of feature XXX, which is better than feature KKK, which was designed for synthetic data, but fails on real images, as it is not robust to clutter, and a new inference method based on ZZZ, which is much faster than any existing method, as shown by our experiments.

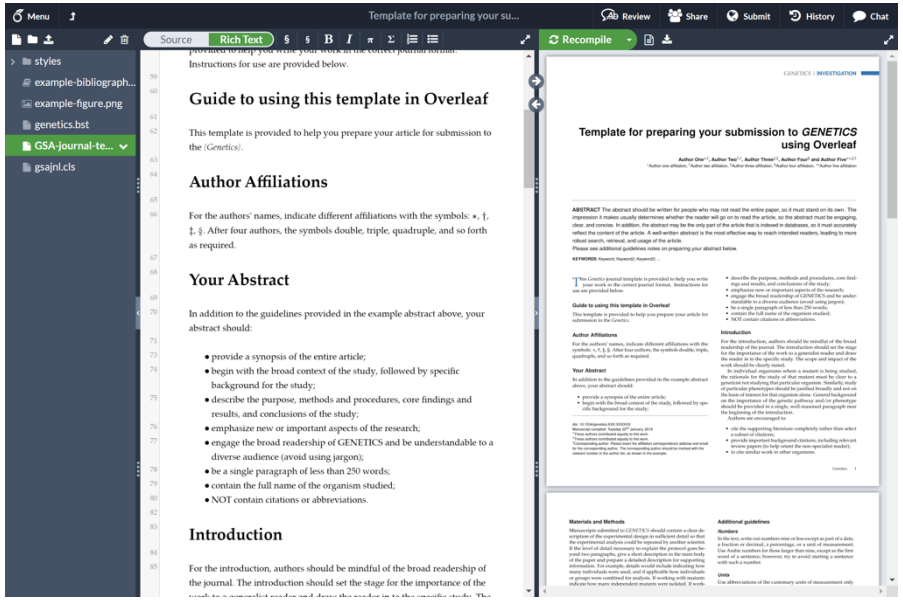
Improved version: The main contributions of our method are the introduction of feature XXX and a new inference method based on ZZZ. XXX proves more robust to clutter in real images than feature KKK [citation], which was designed for synthetic data. ZZZ is an order of magnitude faster than existing inference methods [citations]. Both advantages are proven by experiments (sec.WWW).

Scientific Writing - Overview

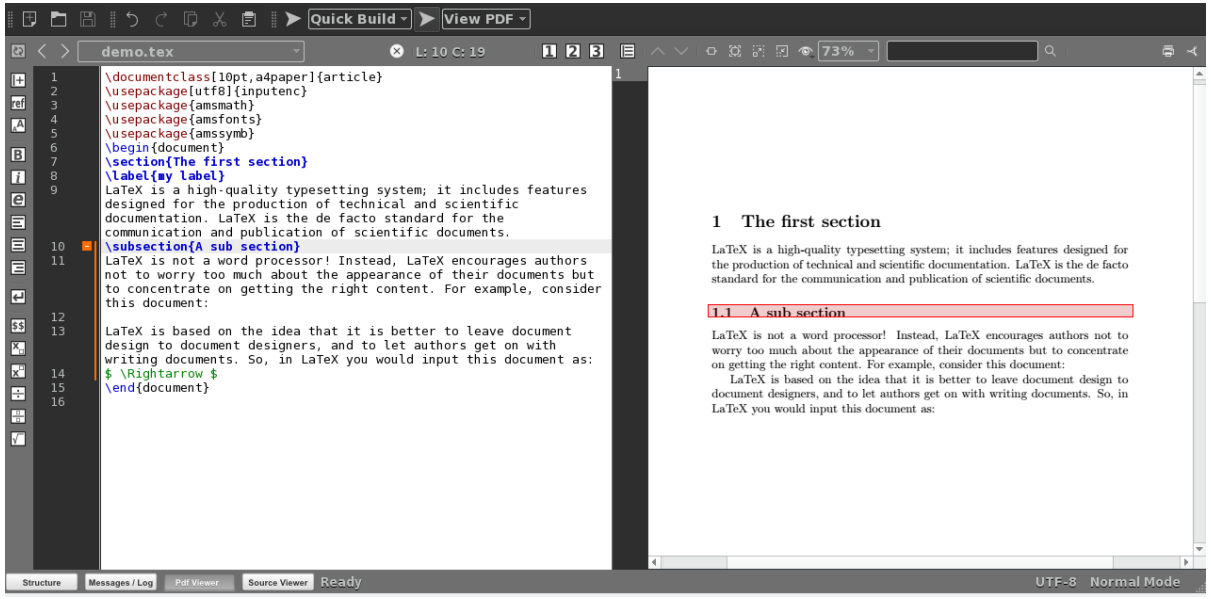
1. Why is it important?
2. Organization – Structure
3. Techniques and Examples
4. **Resources**
5. The fine line between references, citations, and plagiarizing

Resources

LaTeX recommended (alternatively MS Word, LibreOffice, etc.)



Overleaf



Texmaker

[https://www.overleaf.com/ , https://www.xm1math.net/texmaker/]

Figures

It should be easy to read the paper in a big hurry and still learn the main points.

The figures and captions can help tell the story.

So the figure captions should be self-contained and the caption should tell the reader what to notice about the figure

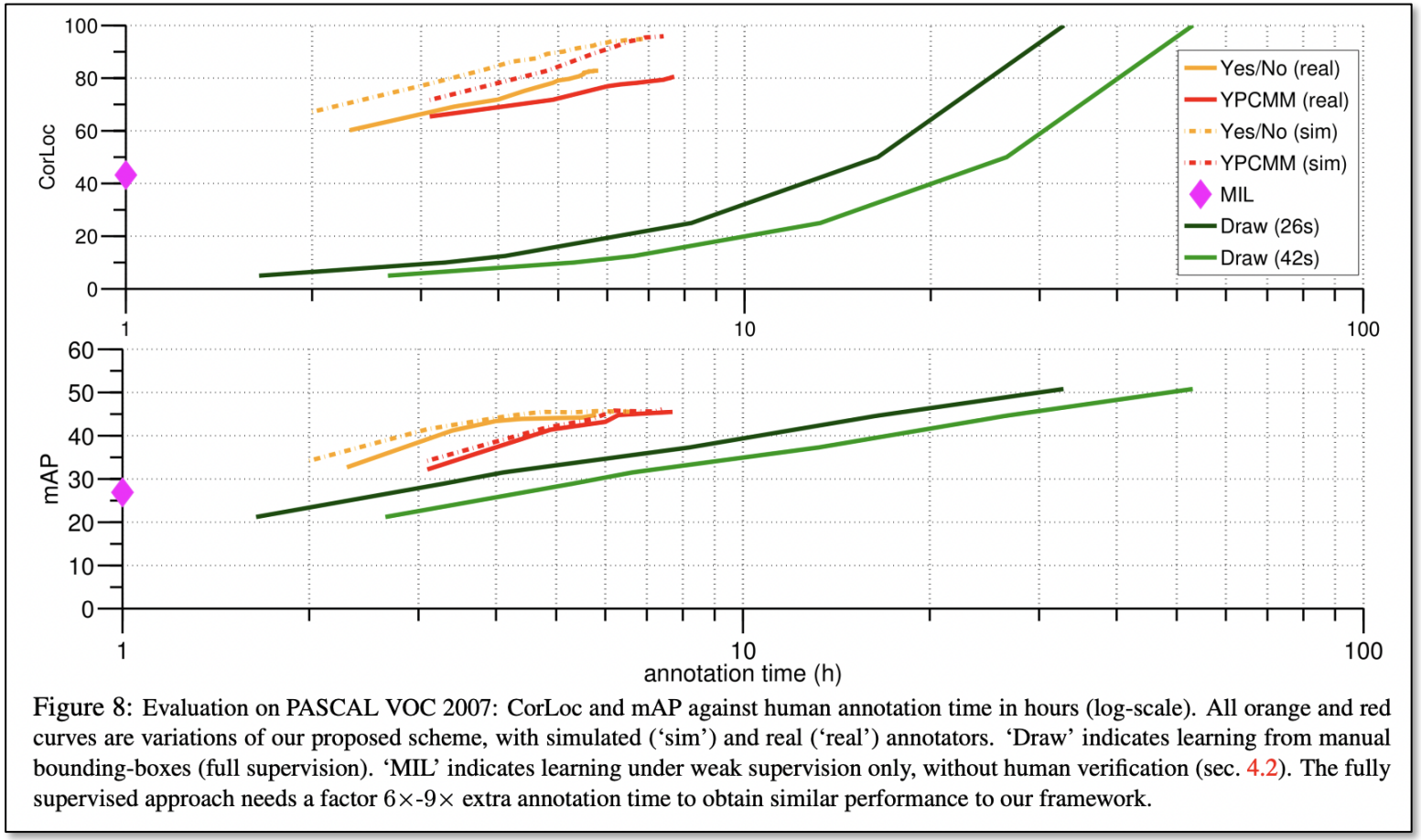


Figure 8: Evaluation on PASCAL VOC 2007: CorLoc and mAP against human annotation time in hours (log-scale). All orange and red curves are variations of our proposed scheme, with simulated ('sim') and real ('real') annotators. 'Draw' indicates learning from manual bounding-boxes (full supervision). 'MIL' indicates learning under weak supervision only, without human verification (sec. 4.2). The fully supervised approach needs a factor 6×-9× extra annotation time to obtain similar performance to our framework.

[Slides from Bill Freeman, How to write a good CVPR submission]

[Figure from Papadopoulos et al, We don't need no bounding-boxes: Training object class detectors using only human verification, CVPR 2016]

Figures

- Use visualization whenever possible
- “A picture is a thousands words”
- Keep figures simple and uncluttered
- Use substantial captions
- Captions should reinforce and augment the text, not repeat it
- Prefer graphs over tables

Figures

Vector graphics vs Raster graphics



[https://cdn.firespring.com/images/e990a5e9-5c64-4a26-b6b2-3b603999c910.jpg]

Figures

Vector graphics vs Raster graphics

	Vector	Raster
Comprised of	multiple reference points and curves	pixels
Scalability	scalable without quality loss	loses quality when scaled
Convertibility	convertible to raster	not convertible to vector
File Formats	EPS, AI,SVG, PDF	BMP, JPG, GIF, PNG

[https://amadine.com/assets/img/articles/vector-graphics/table-of-comparison-of-vector-and-raster-graphics.png]

Scientific Writing - Overview

1. Why is it important?
2. Organization - Structure
3. Resources
4. Techniques and Examples
5. **The fine line between references, citations, and plagiarizing**

How to avoid plagiarism

- Science builds on **existing knowledge**

- **Use references**
  - Describes where information is from
  - Acknowledged sources vouch for accuracy
  - Gives credit to the authors of the original description

- **Direct transcription**
  - Make it clear that it is a quotation – use quotation marks
  - Inform of the source
  - Use references – credit to the authors

- Never plagiarize – it will be discovered and you will be branded as a swindle

Checklist

Can a reader with the right background:

- understand the article
  (Sentences are connected – logical structure)
- follow math equations and proofs
  (Define symbols and notation and use them consistently)
- replicate experiments
- find all assumptions in the text
- find all references and citations
- not find any spelling or grammar errors
- not fall asleep ;)

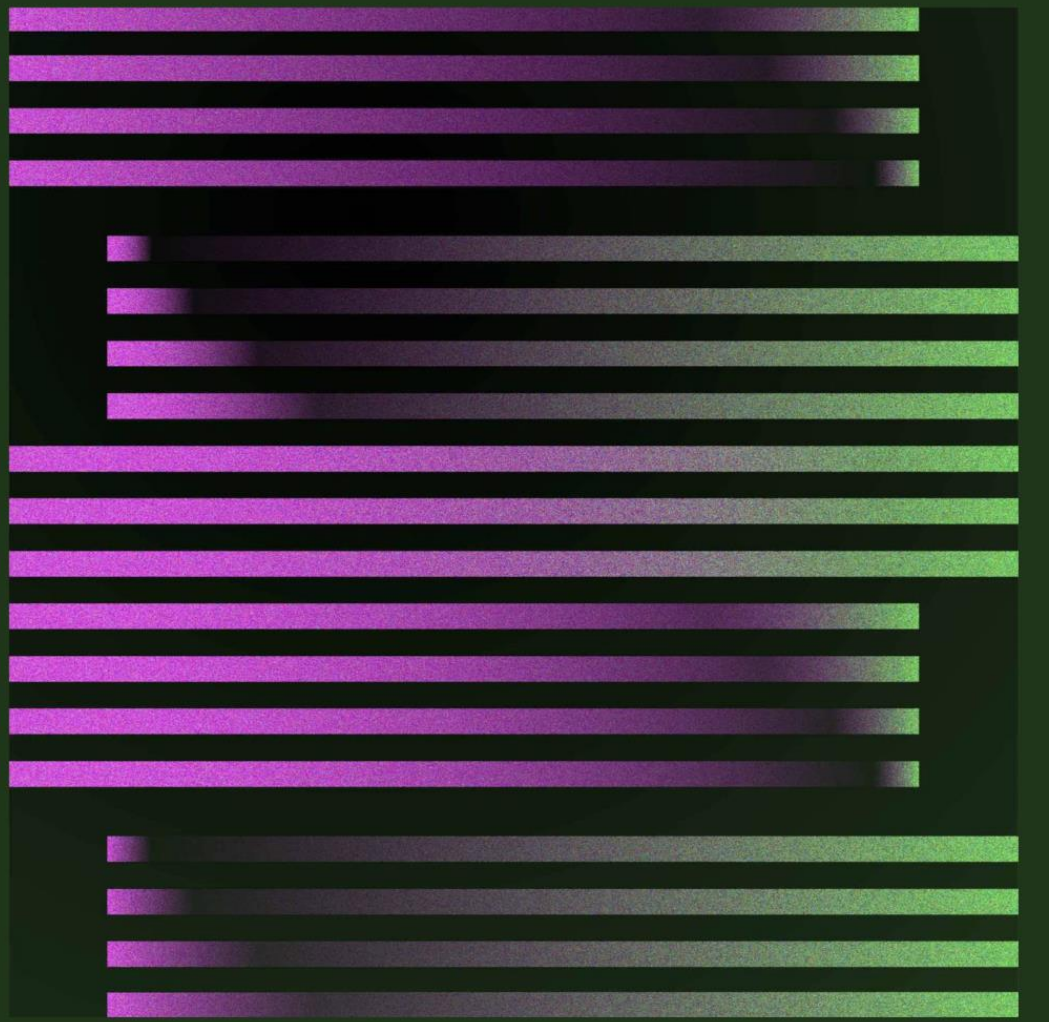
OpenAI

API RESEARCH BLOG ABOUT

ChatGPT: Optimizing Language Models for Dialogue

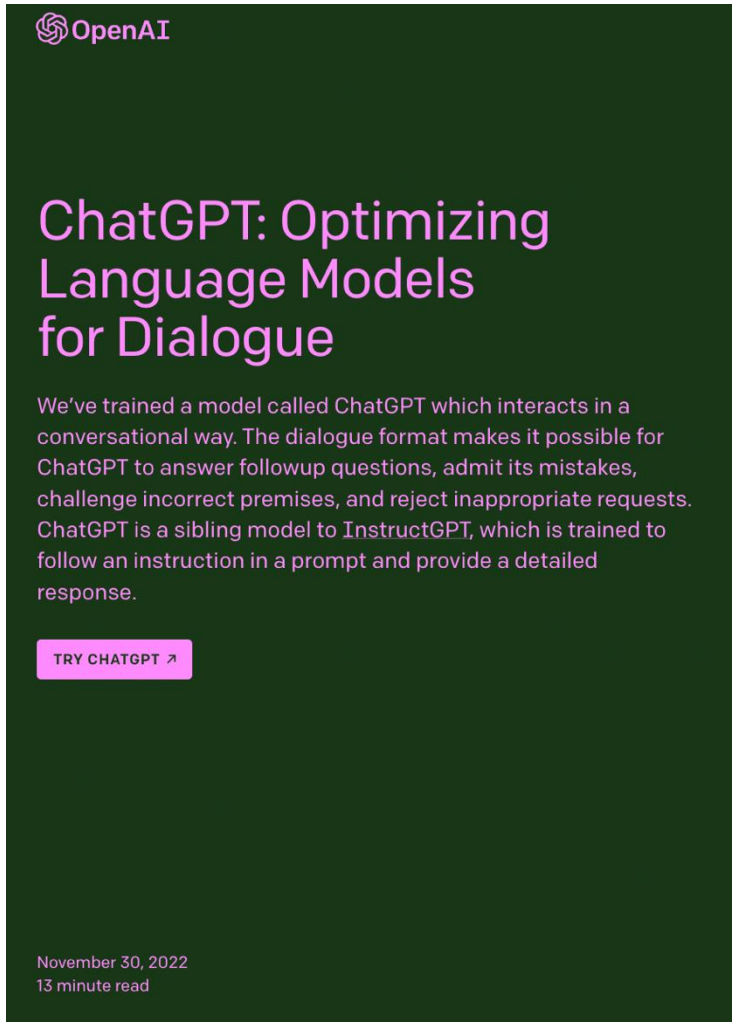
We've trained a model called ChatGPT which interacts in a conversational way. The dialogue format makes it possible for ChatGPT to answer followup questions, admit its mistakes, challenge incorrect premises, and reject inappropriate requests. ChatGPT is a sibling model to InstructGPT, which is trained to follow an instruction in a prompt and provide a detailed response.

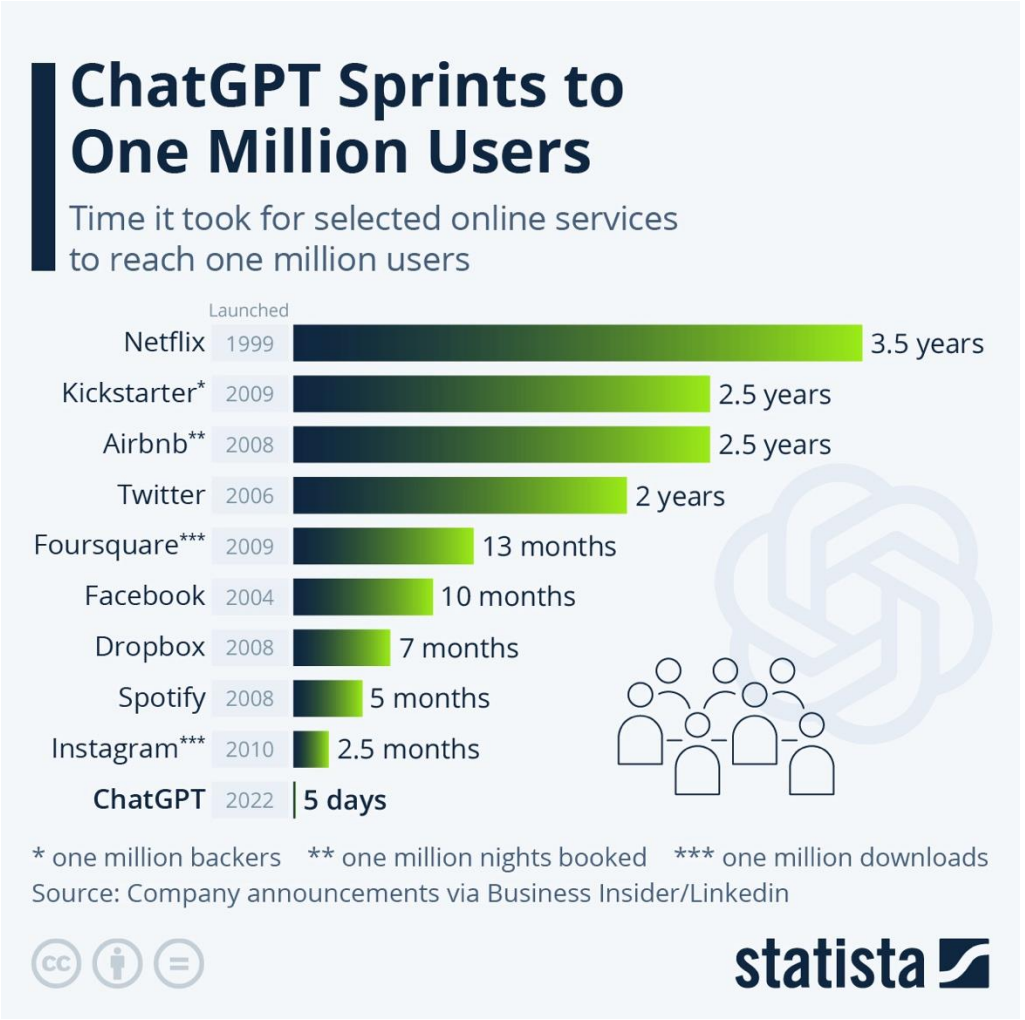
TRY CHATGPT ↗



November 30, 2022
13 minute read

ChatGPT


OpenAI
ChatGPT: Optimizing Language Models for Dialogue
We've trained a model called ChatGPT which interacts in a conversational way. The dialogue format makes it possible for ChatGPT to answer followup questions, admit its mistakes, challenge incorrect premises, and reject inappropriate requests. ChatGPT is a sibling model to InstructGPT, which is trained to follow an instruction in a prompt and provide a detailed response.
TRY CHATGPT ↗
November 30, 2022
13 minute read


ChatGPT Sprints to One Million Users
Time it took for selected online services to reach one million users
Launched
Netflix 1999 3.5 years
Kickstarter* 2009 2.5 years
Airbnb** 2008 2.5 years
Twitter 2006 2 years
Foursquare*** 2009 13 months
Facebook 2004 10 months
Dropbox 2008 7 months
Spotify 2008 5 months
Instagram*** 2010 2.5 months
ChatGPT 2022 5 days
* one million backers ** one million nights booked *** one million downloads
Source: Company announcements via Business Insider/Linkedin
statista

D

write a few tips about scientific writing

1. Clearly define your research question or hypothesis.
2. Use clear and concise language, avoiding unnecessary jargon.
3. Organize your writing into logical sections with informative headings.
4. Provide sufficient background information and context to help readers understand the significance of your work.
5. Use data and evidence to support your arguments, and make sure to explain how you collected and analyzed your data.
6. Use tables and figures to effectively communicate your results.
7. Clearly state your conclusions and their implications for future research.
8. Proofread your writing carefully for grammar, spelling, and punctuation errors.
9. Follow the style and formatting guidelines of the journal or publication you are submitting to.
10. Be open to feedback and criticism, and use it to improve your writing.

D

what is mathematical modeling?

Mathematical modeling is the process of creating a mathematical representation of a real-world system, phenomenon, or process. The model can be used to analyze, predict, and understand the behavior of the system, and can help in making decisions, designing experiments, and developing new theories or technologies. Mathematical models can take many forms, including equations, graphs, simulations, and other mathematical structures.

Generative AI

Discussion of the need to use AI as a tool to learn, not just to produce content

Plagiarism

Who is responsible?

Be aware of the limits of ChatGPT

Properly report in case of usage

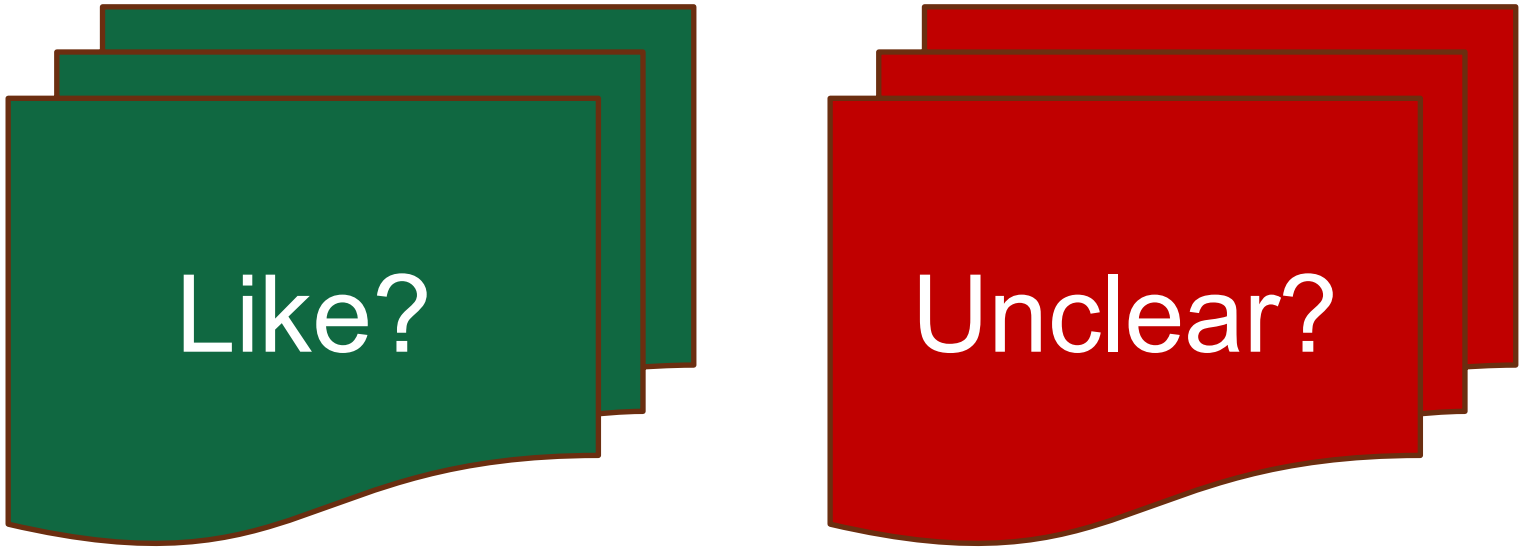
References

- Vetterli, How to write a (hopefully good) paper, http://mri.beckman.illinois.edu/resources/good_paper.pdf
- Freeman, How to write a good CVPR submission, https://billf.mit.edu/sites/default/files/documents/cvprPapers.pdf
- Durand, Notes on writing, http://people.csail.mit.edu/fredo/PUBLI/writing.pdf
- Bertsekas, Ten simple rules for mathematic writing, https://www.mit.edu/~dimitrib/Ten_Rules.pdf
- Ernst, How to write a technical paper or a research paper, https://homes.cs.washington.edu/~mernst/advice/write-technical-paper.html
- Knuth, Notes on technical writing, http://www.ai.mit.edu/courses/6.899/papers/knuthAll.pdf
- Cobb, Scientific writing, Stanford HRP 214, https://web.stanford.edu/~kcobb/writing/

References

- Shewchuck, Three sins of authors in computer science and math, http://www.cs.cmu.edu/~jrs/sins.html
- Memrin, What's wrong with these equations, http://www.ai.mit.edu/courses/6.899/papers/mermin.pdf
- Kajiya, How to Get Your SIGGRAPH Paper Rejected, https://www.siggraph.org/sites/default/files/kajiya.pdf
- Nature, Scitable, Scientific papers, https://www.nature.com/scitable/topicpage/scientific-papers-13815490/
- Gunning, The technique of clear writing
- Halmos, How to write mathematics

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